NORTHSTAR MANUFACTURING UK

Full portfolio reference

Twelve-phase narrative and learning pack, followed by detailed requirements, stories, tests, evidence, data definitions and control catalogues.

Handover: 12 October 2026 · Synthetic baseline: 11 October 2026 Independent simulated portfolio · Free-tool route

This PDF is a reading reference. The repository contains the executable scripts, complete raw synthetic records and editable formats. CSVs are authoritative tabular artifacts; no real records, stakeholder approvals or SAP test results are fabricated.

Contents: project orientation; phases 1–12 and learning reviews; maintained registers; quality gate; process diagrams; catalogue appendices. Use your PDF reader’s bookmarks sidebar to jump to a section. Root README gives the file index.

What was built — NorthStar project guide

Independent simulated portfolio. Initial build 11 October 2026; handover refreshed 12 October 2026. This guide describes actual artifacts and clearly separates the running portfolio application from a future SAP implementation. The synthetic data snapshot remains 11 October 2026.

1. The direct answer

Yes: the twelve-phase **artifact pack** and a **read-only, interactive portfolio application** have been built. You can run the application locally, inspect records and analysis, read the documents, and publish the static application using GitHub Pages after uploading the repository and enabling Pages.

No: this is not a working SAP ERP implementation or a transactional order-management backend. Creating orders, reserving stock, enforcing credit approvals, posting goods issue, issuing legally valid invoices and clearing real payments are requirements/design scenarios in the portfolio. They are not executable business transactions in this app.

The earlier phrase “full project built” referred to the artifact pack and viewer. It did not mean a production ERP system, completed SAP UAT or verified learner mastery. Those distinctions are essential to honest portfolio use.

2. What you can actually do in the application

Screen	Implemented behaviour	Boundary
Overview	Six calculated synthetic KPI cards, exclusions and representative order link	Snapshot, not live business performance
Orders	Search by order/customer/issue; status filter; export filtered CSV; open linked delivery/invoice/allocation detail	Read-only records; no order creation or modification
Process maps	Switch between AS-IS and TO-BE SVG flowcharts	Proposed process logic, not a workflow execution engine
Requirements	Search 23 requirements and inspect acceptance, dependencies and trace links	Proposed requirements, not stakeholder-approved scope
Data & KPIs	Inspect a chart, formulas, populations and limitations	Generated snapshot; filters on the order list do not recalculate cards
UAT & checks	Expand 22 test designs with prerequisites, steps and expected results	Every SAP case remains Not executed
SAP evidence	Read nine source-register entries and open official links	Seven reviewed sources, two unverified pages; no tenant demonstration
Document library	Select embedded phase documents and download PDF/workbook/presentation/setup files	Download files require the complete project folder or hosted repository

There are no logins, roles enforced by the app, live SAP APIs, cloud databases, payment services or secret keys. The role-access matrix describes a proposed business control; it is not implemented authentication. The SQLite database is an analytical output for local inspection, not a backend used by the browser.

3. What work was done

- Converted the supplied scope into a charter, representative order, learning plan and maintained registers.

- Created explicitly simulated stakeholder discovery and mapped conflicting needs to proposed policies.

- Built AS-IS and TO-BE flowcharts with ownership, handoffs and exceptions; separated hypotheses from proven causes.

- Authored 3 business, 15 functional and 5 non-functional requirements, 12 business rules, 15 stories and 22 UAT cases; linked them using stable IDs.

- Reviewed official SAP sources and pinned the reference to S/4HANA Cloud Public Edition 2602; recorded access dates and uncertainty.

- Built a reproducible 300-order synthetic dataset with separate customer, product, order, delivery, invoice, receipt, allocation and event tables.

- Implemented six KPI calculations in Python and independently in SQLite; executed 24 data checks and a hand-calculated fixture.

- Built a ten-risk register, role change/training/support plan and illustrative option comparison that separates capacity from cash savings.

- Created an offline HTML/CSS/JavaScript portfolio viewer, editable workbook and report, ten-slide presentation, case study and conditional interview/resume pack.

- Added local serving, root website entry, GitHub Pages preparation, real-browser checks, screenshot evidence and PDF handover materials.

AI performed the drafting, coding and recorded tool execution in this conversation. Your personal contribution so far is supplying the brief, scope, background and preferences. You have not yet demonstrated independent review or reproduction here. Update that statement as you do the practical tasks; do not claim personal implementation/testing work merely because artifacts exist.

4. All twelve phases and their outputs

Phase	What is present	What remains
1 Charter	Scope, objectives, assumptions, decisions, schedule, NS-ORD-1001	Tailoring to actual study time and primary target role
2 Discovery	Stakeholder matrix, questions, role-play, findings and RACI	Real stakeholder engagement
3 AS-IS	Flowchart, eight-step register and root-cause hypotheses	Validate current process and causes with evidence
4 Requirements	BRD, 23 requirements, 12 rules, access and traceability	Business review and policy approval
5 TO-BE/SAP	Future flow, controls, data needs and source register	Tenant-specific feasibility and settings
6 Fit-gap	Fifteen functional desk assessments plus NFR discussion	System demonstrations against criteria
7 Backlog	Fifteen stories with criteria, dependencies and exceptions	Delivery planning and implementation
8 Analysis	300 orders, dictionary/model, six KPIs, SQL, checks and charts	Broader real data and baseline validation
9 UAT	Twenty-two cases, criteria/severity and design walkthrough	SAP execution and user acceptance
10 Change/case	Ten risks, role impacts, training/support and scenario costs	Agreed operating model and business case
11 Packaging	README, viewer, editable files, deck and case study	Optional GitHub publication by you
12 Coaching	Conditional project descriptions, interview scripts and questions	Your answers, feedback and mastery review

5. How the project works technically

The source files are Markdown documents, CSV catalogues, authored Python, JavaScript and CSS. Python generates deterministic synthetic tables and calculations, then builds the viewer by embedding a JSON snapshot and the documents into HTML. That file runs in the browser without a backend. The same static files can be served locally or by GitHub Pages.

Flow: authored catalogues and generator → CSV / SQLite / calculated results → generated HTML and editable reports → browser exploration. No browser action writes to the original CSVs. Editing the workbook alone does not update the viewer. To change its source data or generated documents consistently, modify source inputs and run the build scripts again.

Root index.html, viewer/index.html and downloads/NorthStar_Offline_Portfolio.html are generated copies of the same app. The root copy makes local and GitHub Pages entry straightforward. Source JavaScript and CSS stay in viewer/app.js and viewer/style.css. This intentional duplication supports a portable single-file app as well as an editable repository.

6. Data and KPI interpretation

The dataset contains 300 orders, 285 deliveries, 262 invoices, 209 receipts and 197 allocations. It includes split deliveries, partial payments, open orders and unmatched receipts. Dates are synthetic UK business-local calendar dates with a snapshot of 11 October 2026. Amounts use integer pence; invoices and receipts are net-only and omit VAT. One line per order is a teaching simplification.

The six full-cohort results are: cycle time 6.16 calendar days; on-time delivery 49.07%; quantity fulfilment 84.89%; billing cycle 1.95 calendar days; order error rate 22.33%; manual-touch rate 76.33%. These are properties of generated data, not actual NorthStar results or achieved improvements.

Cycle time includes 232 fully received orders and excludes 51 open noncancelled orders. On-time delivery counts 132 successes among 269 due initial promises, including overdue open failures. One unknown promise is separately disclosed. Billing cycle covers 262 invoiced deliveries and excludes 22 goods-issued unbilled deliveries. Read the KPI contract before interpreting any number.

The model aggregates child records before combining order measures. A delivery × payment join can otherwise multiply values. A partial payment's analytical remaining balance is not proof that an SAP invoice has been cleared.

7. How to run it locally

Extract the repository ZIP, open a terminal in the northstar-otc folder and run `python scripts/serve.py --open`. Visit `http://127.0.0.1:8000/` if it does not open. Windows can use `py`; macOS/Linux may use `python3`. The server needs only Python's standard library and stops with Ctrl+C.

Double-clicking index.html is also an intended offline path for browsers allowing file URLs. This environment's browser policy blocks that protocol, so actual verification used local HTTP. Refer to RUN_LOCALLY.md for alternatives, the five-minute tour, rebuild commands and troubleshooting.

8. GitHub access and the free route

Push the complete folder contents to a new public repository. Keep index.html at the root. Enable Settings → Pages → Deploy from a branch → main → /(root). After deployment, use the actual URL GitHub displays. A typical project URL has the form https://YOUR_USERNAME.github.io/northstar-otc/; this is a format example, not an existing deployed site.

GitHub Pages can host this static project from a public repository on GitHub Free. The one-time Pages setup is needed; pushing alone is insufficient. GITHUB_SETUP.md includes exact commands, GitHub Desktop guidance and official sources accessed on 11 October 2026. No custom domain or paid service is required for this planned route. Existing device, internet and chat/cloud costs are separate.

Nothing has been pushed to your account or deployed by this handover. GitHub hosting must be checked after your own deployment.

9. Verification and screenshot meaning

Python/SQLite data checks and catalogue consistency checks are recorded in quality and phase-08/data. Node smoke checks verify viewer logic. Real Chromium UI checks, when present, are recorded separately in quality/browser-checks.json with the tested HTML hash, browser version and screenshot references. Screenshots in the handover PDF come from that running local portfolio, not the earlier generated concept image and not SAP software.

The browser exercise covers navigation, search, filters, CSV download, record detail, requirement links, UAT expansion, process switching, evidence, document selection, download links, repository-prefix paths and a mobile viewport. It is not a comprehensive security/accessibility certification or real SAP UAT. No real business users were interviewed or tested. LibreOffice visual rendering is not verified.

10. What “complete” means here

The twelve-phase artifact pack and defined read-only viewer are built. Their exact execution checks and remaining limits are reported, rather than calling them production ready. The original final learning gate stays open until you can explain the decisions, calculations and your personal contribution independently. The files do not prove a live SAP implementation or guarantee employment.

Turning this into a transactional ERP application would be a separate software-development scope: persistence, authenticated roles, transaction rules, inventory/reservation logic, accounting semantics, integrations, security, operational support and system acceptance. That work is neither required for this BA portfolio nor silently claimed as completed.

11. Where to start learning

Read Phase 1 and explain why 60 on hand cannot confirm 100 requested. Follow FR-004 through its story and tests. Recalculate the four-order fixture. Explain physical dispatch versus posting versus receipt. Then work through the short task and three questions in each phase. Resume wording should evolve only as you personally review, change, reproduce and explain the work.

12. Handover files

The Project Guide PDF explains the build, real screenshots, local use and GitHub setup. The Full Portfolio PDF supplies the narrative documents and detailed catalogue appendices. Editable originals and raw synthetic data remain in the repository because PDFs are not a suitable replacement for executable scripts or tabular source data. All formats are readable with free tools.

Phase 1 — Project charter

Supersession note — 11 October 2026: the user authorized building all phases. This Phase 1 draft is retained as the starting charter. Current status and decisions are in ../registers/progress.md and ../registers/decisions.md. Public Edition 2602 is now the research reference. Initial “not started” and evidence-pending statements below describe 8 October, not the current artifact set.

Version 0.1 | 8 October 2026 | Independent simulation | Draft; learner review pending

Executive summary

NorthStar Manufacturing UK is a fictional company selling stocked manufactured products to domestic business customers. The project examines poor order visibility, stock-related fulfilment delays and late billing caused by manual handoffs.

The work will describe the current process, investigate potential causes, define requirements and propose a future process informed by official SAP sources. Synthetic analysis and honest validation records will support a portfolio for junior Business Analyst, ERP Business Analyst and Process Analyst roles. The primary target role is unresolved.

Order-to-Cash means the journey from receiving a customer order through collecting and matching its payment. The project demonstrates analysis, not production readiness or an implemented transformation.

Business problems

Problems are supplied scenario facts. Consequences and causes below are hypotheses, not observed findings.

ID	Problem	Potential operational consequences	Evidence needed in a real business
PROB-001	Poor order visibility	Repeated status enquiries; delayed orders overlooked; unclear ownership	Enquiry logs, order histories, ageing reports
PROB-002	Stock-related fulfilment delays	Missed promises; partial deliveries; replanning effort	Stock commitments, promised dates, delivered quantities
PROB-003	Late billing caused by manual handoffs	Dispatched orders wait for billing; payment period starts later	Dispatch evidence, posting timestamps, invoice dates, handoff records

An overdue invoice is a symptom. Unclear billing ownership is a possible cause requiring evidence. Simulated discovery will not be presented as actual stakeholder research.

Objectives and proposed business measures

Objective	Candidate measures	Business target
Make order progress and exception ownership visible	Manual-touch rate; ability to identify open order status and owner	Unresolved
Improve stock-related fulfilment decisions	On-time delivery percentage; order fulfilment rate	Unresolved
Reduce avoidable billing delays	Billing cycle time	Unresolved
Improve order quality and flow	Order error rate; order cycle time	Unresolved

KPI formulas, date bases, denominators and exclusions will be agreed before analysis. Targets require a baseline, customer commitments, capacity and cost assumptions. Synthetic results can inform illustrative targets but cannot establish real performance or achieved improvements.

Scope

- One fictional company and one warehouse; GBP reporting currency.

- Domestic business customers buying stocked manufactured products.

- Customer order → sales order validation → availability check → fulfilment → delivery and goods issue → billing → payment allocation → reporting.

- Sales, Warehouse, Finance, Customer Service and IT stakeholders.

- Data quality, access permissions, auditability, reporting timeliness, usability and process ownership where relevant.

- Exceptions: missing/incorrect customer data; insufficient stock; partial delivery; credit/order block; order cancellation; incorrect invoice; partial/unmatched payment.

Excluded initially: manufacturing planning, procurement redesign, international trade, complex tax configuration, full migration and production deployment. Record dependencies rather than silently expanding scope. Basic invoice/tax-data needs remain relevant.

Physical events and system postings

A physical event happens in the operation. A system posting records an event or its accounting consequences. Picking removes goods from storage; dispatch sends goods out of the warehouse; customer receipt occurs when the customer receives them. A goods-issue posting records an outbound stock movement in the system. Billing creates an invoice; payment allocation matches received money to an invoice or account.

These events can occur at different times. A goods-issue record alone does not prove customer receipt. Later KPI definitions must distinguish dispatch from receipt. These are introductory business definitions; exact SAP behaviour requires edition-specific verification.

Stakeholder overview

These are proposed fictional responsibilities, not interviewed people or recorded approvals.

Role	Proposed responsibility	Interest
Sales	Validate commercial order information; communicate delivery commitments	Reliable promises and straightforward processing

Role	Proposed responsibility	Interest
Warehouse	Pick, pack, dispatch and record fulfilment information	Clear priorities and accurate quantities
Finance	Credit-control policy, billing oversight and payment allocation	Correct invoices, controlled exposure and traceable payments
Customer Service	Status communication and coordination of customer exceptions	Consistent information and escalation routes
IT	Access, feasibility, integration and support assessment	Maintainable solutions
Learner as portfolio analyst	Produce, review and explain analysis and validation designs	Credible evidence of understanding

Detailed billing ownership and handoff rules remain unresolved. No sponsor approval is claimed.

Reference model and choices

Provisionally select SAP S/4HANA Cloud Public Edition to keep standard-process research within a defined product boundary. An edition identifies a product offering/operating model; a release identifies its version. The reference release is unresolved. If learner access uses another edition, assess whether to change the reference before making capability claims.

This selection is a learning decision, not an investment recommendation. No edition-specific capability has yet been verified or demonstrated. Record official SAP source URLs and actual access dates before using claims in the future-state design. Do not substitute another edition's documentation without flagging the difference.

Alternatives to assess later include improving manual controls, standardising the process, and an SAP-aligned future state. Prefer standard processes when they meet needs; justify any extension and its maintenance cost.

Constraints, dependencies and unresolved choices

The user requires a free working route. No paid SAP environment, Office licence, course, domain or hosting is required. Use documentation and synthetic analysis, with free editing and analysis tools. See free-working-route.md for cost boundaries and evidence limitations.

See the linked register (../registers/assumptions.md). SAP access, study time and primary target role are pending. Baselines, business targets, current tools, partial-delivery policy, delivery promises, billing ownership and payment exception rules remain provisional or unresolved.

Portfolio completion criteria

- All three problems trace to relevant requirements, process changes and validation evidence.

- Normal flow and all seven exceptions are covered consistently across maps, stories and tests.

- SAP claims have official sources and access dates or explicit uncertainty; documentation and demonstrations are distinguished.

- Synthetic data relationships and KPI calculations are reproducible and checked.

- SAP test cases remain Not executed unless run; walkthrough issues are design issues, not executed SAP defects.

- Benefits and financial scenarios are labelled; capacity released is distinguished from cash savings.

- Resume claims match actual completed work; learner can explain key decisions and AI use.

These are portfolio criteria, separate from unachieved fictional business targets. Production readiness would require real discovery, configuration, integration, security assessment, system testing, business acceptance and operational preparation beyond this scope.

Deliverables and phase checkpoints appear in the delivery plan (delivery-plan.md).

Learning and delivery plan

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Provisional: 12 weeks at 4–6 hours per week, approximately 48–72 hours. This is a learning estimate, not an implementation commitment. Actual dates and pace await the learner's answer. Do not advance phases before reviewing learner answers unless the learner asks to continue.

Week	Phase and planned deliverable	Completion checkpoint
1	1: Charter and plan	Scope, evidence boundaries and approach understood
2	2: Stakeholders and simulated discovery	Sources labelled; disagreements recorded
3	3: Current-state flow and causes	Owners, handoffs, exceptions and evidence gaps visible
4	4: BRD, requirements and traceability	Requirements justified and testable
5	5: SAP research and future-state design	Choices supported or explicitly uncertain
6	6: Desk-based fit-to-standard assessment	Gaps, alternatives and validation needs recorded
7	7: Focused backlog	12–15 substantive stories with linked acceptance criteria
8–9	8: Synthetic data and KPI analysis	Approximately 200–500 orders; reproducible checks and calculations
10	9: UAT design and available validation	Around 20 meaningful cases; honest execution status
11	10: Risks, change and business case	Approximately 10 substantive risks; transparent option/benefit assumptions
12	11–12: Packaging, interviews and final review	8–10 slide presentation; claims supported; learner explains decisions

Each week: roughly one hour of concepts, two to three hours of deliverable work and one hour of review/explanation practice. Extend the plan if research or learner review needs more time; do not cut evidence checks to meet the estimate.

Phase 1 practical task

Using NS-ORD-1001, write three sentences: identify one possible order-processing error, suggest a control, and name evidence that would show whether the control worked. Use quality-review experience without disclosing confidential employer information.

The three setup questions have already been asked: SAP practice access, available study hours and preferred target role. Answers are pending. After tailoring the charter, ask three understanding-check questions and review the learner's answers before Phase 2. Do not mistake the instruction to use the cloud workspace for an answer about SAP access.

Selected route — no required paid software

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Decision: use documentation, synthetic data and reproducible analysis. SAP practice access is still unknown and is not required to complete the portfolio. Do not buy a subscription or start a payment-card trial for this project.

This route best fits a beginner building evidence of business analysis: the work can demonstrate clear requirements, process reasoning, controls, data analysis and test design without implying hands-on SAP implementation. Public Edition remains a provisional research reference, not a free SAP system or purchase recommendation.

Tools and formats

- Read the complete Phase 1 pack in the standalone HTML file using a browser. It has no external fonts, scripts, accounts or network requests. Use the browser's Print / Save as PDF option if available.

- Edit Markdown sources using a plain-text editor. They are the authoritative versions used to generate the pack.

- Open DOCX and XLSX copies in the free LibreOffice Writer and Calc applications. Microsoft Office is not required. Layout may vary between applications; LibreOffice rendering has not been tested here.

- CSV copies of each workbook sheet provide simple, portable access. CSV does not retain workbook formatting.

- Later data work will use CSV, free Python tools and optionally SQLite. A spreadsheet-based dashboard can use LibreOffice Calc. Power BI and paid sharing services are not required.

- Use publicly accessible official SAP documentation and free SAP Learning material where available. Check access and edition before relying on a source. If a resource requires payment, find another official public source or label the claim unverified. Do not assume practice systems, certifications or all SAP training are free.

- Keep artifacts in the workspace and download local copies. GitHub publishing remains optional and requires a later user request; no paid hosting or domain is needed.

Cost boundary

Required incremental software purchases for this planned route: £0. This is a project tooling decision, not a business-case saving. Existing device, internet and this chat/cloud service are outside that estimate; their pricing and availability are not controlled by this project. No guarantee of free ongoing cloud hosting is made. Downloaded source files can be used independently of this workspace.

Alternatives and limitations

A paid SAP practice environment could add demonstrated system evidence but is not needed for the chosen route. A documentation-based portfolio cannot claim system configuration, integration execution or SAP UAT execution. All unrun SAP cases will remain Not executed. Data checks and walkthroughs will be labelled separately.

The full twelve-phase project is still planned, but only Phase 1 is built. Learning reviews remain part of the approach; requesting a free version does not supply answers or prove understanding.

Phase 1 — Learning review

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Status: not yet answered or assessed. This is a learning exercise, not a stakeholder interview or an executed SAP test.

Purpose

A charter agrees what problem the project investigates, what work belongs in scope and what evidence will count as completion. It prevents a proposed software solution from replacing a clear business need.

Your quality-audit experience is relevant: an error observation identifies a problem; a control addresses risk; evidence shows whether the control operated. A proposed cause still requires investigation.

Worked example

NS-ORD-1001 requests 100 brackets. Only 60 are physically on hand, and their availability for this customer is unresolved. A possible error is promising all 100 for 9 October without checking commitments and restrictions. A proposed control is to validate available quantity and a defensible delivery date before confirming the promise. Evidence would include the availability information used, the recorded promise and who confirmed it. These are proposed controls and required evidence, not verified SAP functions or actual records.

An alternative is offering a partial delivery with customer agreement. That may meet an urgent customer need but add handling and transport effort. We have not decided whether it is the correct policy.

Practical task

Write three sentences proposing a different possible error, a suitable control and evidence you would inspect. You may use missing customer information or a delayed billing handoff. Avoid confidential employer details.

Three understanding-check questions

1. For NS-ORD-1001, why can we not promise 100 units merely because 60 units are physically in the warehouse? Name two facts you would check before confirming a delivery date.

2. Goods leave the warehouse on Monday, the goods-issue posting is recorded on Tuesday, and the customer receives them on Wednesday. Which event would you use to assess a promise of customer receipt, and why must the other dates remain separate?

3. Explain the difference between completing this portfolio and achieving a business improvement. Give one piece of evidence for each and identify which we currently have.

Review criteria: distinguish physical stock from available stock; distinguish dispatch, posting and receipt; distinguish portfolio artifacts from business outcomes. Learner responses and feedback remain pending. No pass or competence claim is recorded.

Representative order — NS-ORD-1001

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All values below are fictional learning assumptions. This is a scenario record, not an SAP transaction or executed test.

Field	Value
Order reference	NS-ORD-1001; portfolio identifier, not SAP-generated
Customer	CUST-001 — Alder Engineering Ltd
Product	PROD-001 — standard mounting bracket
Warehouse	WH-01
Order received	5 October 2026
Quantity ordered	100 units
Unit selling price	£25 excluding VAT
Net order value	100 × £25 = £2,500
Customer-requested receipt date	9 October 2026; not a confirmed promise
Initial stock	60 units physically on hand; usable and uncommitted quantity unresolved
Payment terms	Illustrative: 30 days from invoice date
Open decision	Partial-delivery permission and defensible delivery promise

Do not infer that the 60 units are available for this customer. Reservations, quality restrictions or inaccurate records could affect availability; these possibilities are unverified.

The net value is not profit, cash received or a tax-inclusive invoice total. No acceptance, dispatch, posting, invoice or payment has occurred in a demonstrated system. Keep later exception variants separate from this base scenario.

Stakeholders and simulated discovery

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Purpose and boundary

Find the business need behind each complaint and identify whose policy decisions conflict. All dialogue below is invented role-play, created to practise discovery. There were no stakeholder interviews, workshops, attendance records or approvals.

Sales

Responsibility: Order capture, validation, commercial promise

Interest: Fast acceptance and customer retention

Influence: High: controls promises

Concern: Checks may slow response

Engagement: Review incomplete/short-stock examples and promise rules

Warehouse

Responsibility: Stock accuracy, picking, dispatch and posting evidence

Interest: Feasible priorities and correct quantities

Influence: High: controls fulfilment

Concern: Sales promises exceed usable stock

Engagement: Walk shortages, split deliveries and event recording

Finance

Responsibility: Credit, billing, receivables and controls

Interest: Accurate charges and supported clearing

Influence: High: financial control owner

Concern: Pressure to bypass blocks; late handoffs

Engagement: Review authorization, billing and payment scenarios

Customer Service

Responsibility: Status communication and exception coordination

Interest: Reliable answers and named owner

Influence: Medium: customer voice

Concern: Conflicting statuses and partial shipments

Engagement: Walk customer enquiry and delivery-agreement cases

IT

Responsibility: Feasibility, roles, integration and reporting support

Interest: Standard, maintainable design

Influence: High: technical feasibility

Concern: Custom dashboard scope and unverified integrations

Engagement: Evidence-led fit review and data contract checks

Interview guide to use in a real discovery

Sales: Walk through the last incomplete order. Which fields were missing, where were they captured and what proved correction? What evidence supports your delivery promise? How do you recognize a duplicate customer purchase order? When can cancellation safely stop fulfilment?

Warehouse: Show how usable stock differs from shelf quantity. What happens when picking finds fewer units? How do you authorize split shipments? Who records dispatch, goods issue and receipt evidence, and how are timing differences reconciled?

Finance: What triggers a credit block and who may release it? What proves a delivery is billable? How do you identify missed or duplicate charges? What distinguishes invoice creation, accounting posting and successful customer output? How do you handle short or unidentified receipts?

Customer Service: What information is needed to answer an order enquiry? Which status is authoritative? How is consent to split shipment recorded? How do you escalate an overdue response and notify the customer without promising an unsupported date?

For every answer ask for an example, record, frequency and contrary case. Do not turn an opinion into a quantified baseline.

Role-play workshop — proposed 60-minute agenda

10 minutes: agree boundaries and terminology. 20: walk NS-ORD-1001. 15: debate policy conflicts. 10: agree provisional requirements. 5: list evidence gaps. This agenda has not been run with people.

Sales: “Can we promise all 100 for Friday so the customer stays with us?” Warehouse: “Only 60 are on hand; we have not checked commitments.” Customer Service: “I could ask whether 60 helps, but I need a defensible date for the rest.” Finance: “Agreement to split does not remove a credit hold.” IT: “First confirm the rules; do not commission a custom promise engine.”

Later, Warehouse says billing belongs to Finance once goods leave. Finance says a dispatch email is still needed. The analyst proposes that Warehouse owns accurate dispatch/posting evidence and Finance owns a daily billable/unbilled review. Neither team can treat the handoff as complete solely because an email was sent. This is a proposed resolution, not approval.

Conflicts and proposed resolutions

Conflict	Proposed resolution	Alternative and trade-off	Decision owner
Fast promise vs feasible supply	Confirm only supported quantity/date; explain shortfall	Provisional estimate can be communicated if clearly unconfirmed	Sales with Warehouse
Urgent partial vs single complete shipment	Explicit agreement and cost awareness; retain remainder	Wait for all stock: lower handling effort, potentially worse service	Customer Service with Sales
Sales urgency vs credit control	Finance decides credit release; Sales requests review	Blanket override is rejected because it removes control	Finance
Billing ownership ambiguity	Warehouse owns fulfilment evidence; Finance owns billing queue and reconciliation	Email checklist is a low-cost interim control	Finance

Findings

See discovery-findings.csv for the 11 linked simulated sources. None establish actual frequency or impact. Validate proposed policies with real role owners before implementation.

Phase 2 learning review — Stakeholders and discovery

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Purpose and essential concept

Discovery finds the needs behind requested features. Your audit feedback experience helps you ask for examples rather than accept a general complaint.

Practical task

Rewrite one delivery-promise interview question so it asks for observable evidence.

Three understanding questions

1. Why might Sales and Warehouse disagree about dates?

2. Who should own a credit-release decision?

3. Why is our role-play not interview evidence?

Status: answers not supplied; understanding not assessed. Review before claiming mastery. Building all phases was authorized; this does not substitute for your learning review.

Proposed responsibility matrix

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

A = accountable for the outcome; R = performs work; C = consulted; I = informed. One accountable role per activity. A/R combines responsibility for this small-company simulation, not every named employee. Credit request and release must remain separated.

Customer/order validation

Sales: A/R

Warehouse: I

Finance: C

Customer Service: C

IT: I

Promise and partial agreement

Sales: A/R

Warehouse: C

Finance: I

Customer Service: C

IT: I

Credit release

Sales: C

Warehouse: I

Finance: A/R

Customer Service: I

IT: I

Dispatch and goods issue

Sales: C

Warehouse: A/R

Finance: C

Customer Service: I

IT: I

Billing and corrections

Sales: I

Warehouse: C

Finance: A/R

Customer Service: C

IT: I

Payment allocation

Sales: I

Warehouse: I

Finance: A/R

Customer Service: C

IT: I

Customer exception communication

Sales: C

Warehouse: C

Finance: C

Customer Service: A/R

IT: I

Role/report configuration

Sales: C

Warehouse: C

Finance: C

Customer Service: C

IT: A/R

AS-IS process and root-cause hypotheses

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

This is a **process flowchart, not BPMN**. Tools and responsibilities are fictional current-state assumptions. Shared owner labels show uncertain handoffs that the TO-BE must resolve.

Full process flowchart is provided in the diagram appendix and as editable .mmd/.dot and zoomable .svg files. This is not BPMN.

The full eight-step register is in as-is-step-register.csv. Cancellation after dispatch is an exception routed to controlled review, not a full returns redesign. Unmatched and partial payments remain visible rather than pretending the order is financially complete.

Five Whys — late billing hypothesis

1. Why is billing late? A dispatched delivery is absent from the billing review list.

2. Why absent? Finance has not received or located the handoff email.

3. Why does that matter? The email is treated as the trigger instead of reconciling all eligible deliveries.

4. Why is reconciliation absent? Ownership of the complete billable population is unclear.

5. Why unclear? The cross-team procedure defines individual tasks but no accountable handoff outcome.

This chain is a hypothesis, not a root cause proven by five repetitions. Alternative explanations include pricing incompleteness, goods-issue timing, invoice schedules or accounting blocks. Compare dispatch, posting, invoice and output dates plus exception reasons before attributing delay. A missing receipt timestamp can be an evidence gap rather than a late physical delivery.

Root-cause test plan

Hypothesis	Evidence that supports it	Evidence that could refute it	Proposed response
Fragmented status causes enquiries	Enquiry timestamps coincide with inconsistent team records	Accurate shared records exist but users cannot locate them	Shared definitions or usability training depending on evidence
Promise ignores commitments	Orders promised beyond usable stock at confirmation	Stock was sufficient but damaged during picking	Availability review or warehouse control depending on cause
Email handoff delays billing	Eligible items wait until email arrives	Items were blocked for pricing despite timely email	Queue ownership or pricing correction depending on cause

Phase 3 learning review — Current-state process

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Purpose and essential concept

A process map shows where work and information cross roles. A late invoice is a symptom; unclear handoff ownership is a hypothesis.

Practical task

Use Five Whys to challenge one billing-delay explanation.

Three understanding questions

1. What evidence could disprove our handoff hypothesis?

2. Where does a physical event differ from a posting?

3. Why call this a flowchart rather than BPMN?

Status: answers not supplied; understanding not assessed. Review before claiming mastery. Building all phases was authorized; this does not substitute for your learning review.

Concise business requirements document

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Business need

PROB-001: unclear order status. PROB-002: stock-related delays. PROB-003: billing handoff delays. The goal is controlled, visible processing, not buying SAP by default. Scope, exclusions and NS-ORD-1001 remain as in Phase 1. The source-labelled discovery is simulated; no real stakeholder acceptance exists.

Business requirements

BR-001

Type: Business

Statement: Enable accountable, traceable order and payment status from intake through allocation and reporting.

Rationale: Reduce uncertainty and prevent lost exceptions.

Source: PROB-001; SIM-CS-01; SIM-FIN-04

Priority: Must

Priority_reason: Core visibility problem

Acceptance: Representative normal, blocked, cancelled and partially paid cases expose current stage, owner, remaining quantity/balance and evidence gaps.

Dependencies: Master data; linked documents; agreed role access

Validation: Trace review and planned UAT TC-001;TC-014;TC-017

BR-002

Type: Business

Statement: Base fulfilment commitments and release on feasible supply, customer agreement and financial controls.

Rationale: Avoid unsupported promises and unauthorized shipments.

Source: PROB-002; SIM-WH-01; SIM-FIN-01

Priority: Must

Priority_reason: Core fulfilment/control problem

Acceptance: Stock shortage, partial permission and block cases follow rules; measures KPI-001/002/003 use the agreed definitions; no achieved improvement claimed.

Dependencies: Stock accuracy; credit policy; transit assumptions

Validation: Planned UAT TC-004 through TC-008; synthetic KPI checks

BR-003

Type: Business

Statement: Make eligible fulfilled deliveries visible to Finance for timely, accurate and traceable billing.

Rationale: Remove avoidable handoff waiting without billing incorrectly.

Source: PROB-003; SIM-FIN-02; SIM-FIN-03

Priority: Must

Priority_reason: Core billing problem

Acceptance: Eligible delivery appears in review population without an email; correction/duplicate/error paths retain links and owner; KPI-004 includes unbilled context.

Dependencies: Posting accuracy; billing relevance; Finance ownership

Validation: Planned UAT TC-012;TC-013;TC-022

Functional and non-functional scope

requirements.csv contains 3 business, 15 functional and 5 non-functional requirements with rationale, priority, acceptance, dependencies and validation. business-rules.csv defines 12 stable rules. traceability.csv connects every FR to problem → BR → process change → story → tests → KPI/control. nonfunctional-trace.csv adds NFR coverage. Access permissions are proposed in access-matrix.csv; the offline viewer has no production authorization enforcement.

Must = essential to a core flow/control; Should = useful with a controlled interim process. These priorities do not imply a committed sprint. Thresholds follow business need: exact key/balance reconciliation, zero unauthorized releases in a test set, daily management review freshness, and formative usability by role. No arbitrary speed or improvement target is treated as agreed.

Requirement review decisions

“Real-time visibility” was narrowed to named status fields and a stated refresh time; real-time integration lacks evidence of need. “Automatic billing” was narrowed to complete billable-population review and controlled invoice creation, including failures. “All stock is available” was rejected because commitments and restrictions matter. “Clear every payment automatically” was rejected because unidentified cash needs investigation.

Unresolved items and change control

Retention period, exact SAP role restrictions, carrier receipt capture, correction method, duplicate customer PO policy and refresh architecture need validation. A duplicate customer-PO control is a discovery question, not silently added to the approved scope. Proposed changes must record source, rationale and impacts on process, rules, stories, tests and KPI definitions. No actual approval has been recorded.

Phase 4 learning review — Requirements

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Purpose and essential concept

A requirement states an observable need, not a vague wish. Audit criteria are a useful analogy for acceptance criteria.

Practical task

Rewrite “make orders visible” with fields, role and an observable result.

Three understanding questions

1. How does a rule differ from a requirement?

2. What makes an acceptance criterion testable?

3. Why keep a source for each requirement?

Status: answers not supplied; understanding not assessed. Review before claiming mastery. Building all phases was authorized; this does not substitute for your learning review.

SAP reference and proposed TO-BE

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Reference boundary

Select SAP S/4HANA Cloud Public Edition **2602**, using the official release-specific feature scope document version 5.0 dated 11 June 2026. This is a stable research reference, not a claim to use the newest release or a live tenant. Public Edition suits a standard-first learning exercise; it is not free SAP hosting. Learning pages support concepts but are not pinned to 2602. Feature entitlement, country scope, activation and exact configuration require validation.

Essential concepts

A business partner is the common customer identity with role-specific information (SAP-005). A sales order records the requested commercial transaction. Availability and scheduling help establish a feasible quantity/date; shelf stock alone is insufficient (SAP-002). An outbound delivery organizes shipping; partial-delivery rules need agreement (SAP-003). Goods issue records the outbound stock event, while physical dispatch and receipt are separately evidenced. Billing charges the customer and connects to receivables, the amount owed (SAP-001). Finance then handles payment matching and differences (SAP-006). No step has been demonstrated in an SAP system.

Proposed process flowchart — not BPMN

Full process flowchart is provided in the diagram appendix and as editable .mmd/.dot and zoomable .svg files. This is not BPMN.

Ownership, controls and handoffs

Sales owns valid order/promise information; Customer Service records agreement and communication; Warehouse owns quantities and dispatch/posting evidence; Finance owns credit, billable-population reconciliation, corrections and allocation; IT supports roles, interfaces and reporting. A handoff is complete when the receiving role has a linked, actionable record or a named exception—not simply an email. RULE-001 through RULE-012 apply. PC-001 through PC-015 identify changes in traceability.csv.

The base NS-ORD-1001 remains unresolved: 100 ordered at £25, 60 physically on hand, requested receipt 9 October. A **separate design variant** assumes 60 are usable and the customer agrees: deliver/bill 60 (£1,500 net), keep 40 open; a £900 partial payment leaves £600 analytical exposure on that invoice. This is an illustration, not an executed order or the baseline record's actual history.

Current vs proposed

Current assumption	Proposed change	Why / alternative
Fragmented status lists	Linked order view with reason and owner	Standard views first; controlled shared register if fit unresolved
Shelf-stock promise	Availability/scheduling review and retained original promise	Manual checked promise can be an interim control
Informal split approval	Explicit permission, quantity and remainder	Complete delivery when customer does not agree
Dispatch-email billing trigger	Finance queue and daily reconciliation	Daily manual checklist is a credible low-cost option
Ad hoc payment matching	Supported allocation and visible unmatched queue	Keep on-account investigation; no forced write-off

Data needs beyond the analytical dataset

Customer identity/status and ship-to; product/UOM; quantity and price conditions; requested/initial confirmed dates; check time and block reasons; delivery references and authorized/actual quantities; dispatch/PGI/receipt timestamps; invoice/accounting/output status; corrections; receipt/allocation amounts; actor and change reason. The synthetic dataset is a deliberately smaller analytical model, not an SAP extraction schema or migration mapping.

Receipt capture may need a manual evidence field or carrier integration; this exact fit is unresolved. Returns/reversals, tax settings, replenishment, bank formats and retention policy are dependencies. Do not propose an extension until standard capture and export are tested. Official URLs and actual access date appear in sap-evidence.csv; two inaccessible Help pages are explicitly unverified.

Phase 5 learning review — SAP reference and future state

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Purpose and essential concept

Standard capabilities inform design; they do not prove your policy is configured.

Practical task

Explain NS-ORD-1001 using order, delivery, goods issue and invoice documents.

Three understanding questions

1. Does goods issue prove customer receipt?

2. Why can requested and confirmed dates differ?

3. What is documented but not demonstrated here?

Status: answers not supplied; understanding not assessed. Review before claiming mastery. Building all phases was authorized; this does not substitute for your learning review.

Desk-based fit-to-standard assessment

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

fit-gap.csv covers all 15 functional requirements. Supported in principle means relevant standard concepts exist, not that all acceptance criteria are satisfied. Process/configuration change means policy and settings must be demonstrated. Integration/extension candidate means investigate standard capture first. Unresolved means evidence is insufficient.

NFR assessment: NFR-001 uses broad role capability SAP-007 but exact segregation is unvalidated; NFR-002 audit field coverage/retention unresolved; NFR-003 refresh architecture unresolved; NFR-004 requires real users; NFR-005 is demonstrated only on the synthetic data subset. No SAP system acceptance is claimed.

Prioritise demos for blocked release, partial delivery, invoice retries/corrections and event-level receipt capture. An external receipt feed could require interface monitoring, retries, security, data mapping and release regression testing. A custom exception view also creates maintenance and access-control obligations. Prefer a standard report plus a controlled exception register until an evidenced gap justifies more.

The locally built portfolio viewer is a learning presentation, not a proposed SAP production extension or proof of FR implementation.

Phase 6 learning review — Fit-to-standard

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Purpose and essential concept

Fit assessment compares a need with evidence and identifies remaining validation.

Practical task

Choose one unresolved fit and propose a standard-first experiment.

Three understanding questions

1. Why is a broad feature not proof of exact fit?

2. When would an extension be justified?

3. What maintenance cost can an extension create?

Status: answers not supplied; understanding not assessed. Review before claiming mastery. Building all phases was authorized; this does not substitute for your learning review.

Focused product backlog

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

backlog.csv contains 15 substantive proposed stories grouped by capability. Each names a role, business value, linked requirement, Given/When/Then criteria, priority, dependencies and exception coverage. Specific fixtures and actions are in the linked UAT cases rather than duplicating long steps in each story.

Suggested sequencing: US-001/003/005/014 establish input and controls; US-004/006/007 handle fulfilment and cancellation; US-008/009/010 handle billing; US-011 covers receipts; US-002/012/013/015 depend on reliable linked information. This is dependency reasoning, not a sprint commitment. No velocity or delivered SAP increment is claimed.

Analysis tasks are separate: AT-001 verify exact credit-release roles; AT-002 define correction route; AT-003 validate receipt capture; AT-004 test reporting extraction/grain; AT-005 conduct actual usability review. They are evidence-gathering tasks, not user-facing product stories.

Phase 7 learning review — Backlog

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Purpose and essential concept

Stories express a role, need and value; acceptance criteria define observable outcomes.

Practical task

Write an extra negative acceptance criterion for partial delivery.

Three understanding questions

1. Which dependencies affect US-004?

2. Is researching an SAP page a product story?

3. Why does a backlog not prove delivery experience?

Status: answers not supplied; understanding not assessed. Review before claiming mastery. Building all phases was authorized; this does not substitute for your learning review.

Synthetic data model, analysis and reproducibility

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Model before generation

Table	Grain	Key	Relationship
customers	One fictional customer	customer_id	None
products	One stocked product	product_id	None
orders	One single-line order	order_id	customer_id → customers; product_id → products
deliveries	One shipment for an order	delivery_id	order_id → orders
invoices	One active net-only invoice per delivery	invoice_id	delivery_id → deliveries
receipts	One incoming net-only receipt	receipt_id	customer_id → customers
allocations	One receipt-to-invoice allocation	allocation_id	receipt_id → receipts; invoice_id → invoices
events	One intake/quality/manual/cancellation event	event_id	order_id → orders

All records are synthetic, generated with Python random seed 20261011. 300 orders, 20 customers, five products and one warehouse; single line per order is a deliberate simplification. A future multi-line model needs order_items and line-level delivery/invoice links. Receipts and invoices can have multiple allocation rows structurally, although this generator mainly demonstrates split payments per invoice plus unmatched receipts rather than every possible many-to-many pattern.

Generation uses an intake cohort from 1 July through 11 October 2026, snapshot 11 October 2026. Orders received near the cutoff have less time to finish; open and future-due counts must be shown. Dates are UK business-local calendar dates; weekends are not excluded. NS-ORD-1001 preserves the Phase 1 facts and remains unconfirmed/unfulfilled. Other distributions are designed teaching examples, not calibrated observations. All financial amounts are **net-only integer pence**, with VAT deliberately excluded from invoices and receipts; this is not a complete UK receivables ledger.

Generation assumptions

Every seventeenth non-base order is cancelled before dispatch. Every nineteenth is left without fulfilment. Some permitted orders split into two shipments; some remain partial. Stock-shortage labels add delay. Billing-handoff labels add billing delay. Errors and manual touches are explicit event records. Pending future events are omitted, not recorded as actuals. No returns, invoice corrections, overpayments, discounts, tax or bad debt are generated; those scenarios remain in design/UAT. This bias intentionally illustrates the problems and cannot prove causation.

Relationships and double-counting

orders 1:N deliveries; deliveries 1:1 invoices in this simplified dataset; invoices 1:N allocations and receipts 1:N allocations. Aggregate delivery quantity to order, allocations to invoice/receipt, and event flags to order before joining. Never SUM order value after joining raw deliveries and allocations. The four-order fixture demonstrates that £10,000 remains £10,000 regardless of child-row count.

Executed baseline results — not achieved improvements

KPI-001

name: Order cycle time

value: 6.1595

unit: calendar days

denominator: 232

basis: Synthetic baseline; not measured business performance

KPI-002

name: On-time delivery

value: 49.0706

unit: %

denominator: 269

basis: Synthetic baseline; not measured business performance

KPI-003

name: Order fulfilment rate

value: 84.8933

unit: %

denominator: 14530

basis: Synthetic baseline; not measured business performance

KPI-004

name: Billing cycle time

value: 1.9504

unit: calendar days

denominator: 262

basis: Synthetic baseline; not measured business performance

KPI-005

name: Order error rate

value: 22.3333

unit: %

denominator: 300

basis: Synthetic baseline; not measured business performance

KPI-006

name: Manual-touch rate

value: 76.3333

unit: %

denominator: 300

basis: Synthetic baseline; not measured business performance

Context: 17 cancelled; 232 fully received; 51 open noncancelled; 1 unknown initial promise; 21 orders with multiple deliveries; 21 partially received; 20 partially paid invoices; 22 goods-issued deliveries still unbilled. These exclusions matter: completed-only cycle and billed-only billing means can look better while unfinished work grows. On-time performance is 132/269 due promises, with overdue open orders counted as failures.

Synthetic invoice balances outstanding total £142,705.00; unmatched cash totals £3,060.00. These are model balances, not real cash or savings. Total order value includes cancelled demand; do not call it revenue. Results describe the generator, not NorthStar operations.

Reproduce and inspect

Run `python scripts/build_data.py` from the project folder; it uses only Python's standard library. CSVs, SQLite database, results JSON, checks and SQL are generated. kpi_queries.sql independently computes the six measures in SQLite. 24 checks passed, including keys, references, date/quantity/amount bounds, known fixture and Python/SQL agreement. SHA-256 hashes in results.json make deterministic reruns comparable. These are executed data checks, not SAP UAT.

LibreOffice route: import each CSV with comma delimiter and UTF-8; retain identifiers as text. Use order_summary.csv as the order-grain table for pivots. It is an exported derived table; source CSV changes require regenerating it. For on-time calculations filter noncancelled known promises due by snapshot; count full receipt by original promise. For billing join invoices to deliveries once at delivery ID. Power BI is optional, not required.

Dashboard design and implementation

The offline viewer shows six cohort-wide KPI cards, denominator/exclusion context, searchable orders with linked shipment/invoice/payment details, requirement traceability, tests and source evidence. Order-list search filters the list only; KPI cards stay clearly labelled as the full snapshot. No role permissions or transactional SAP behaviours are simulated by the viewer.

Future-state scenario boundary

No future-state KPI uplift is presented. Phase 10 models illustrative handling-time capacity separately; it is not subtracted from this baseline or claimed as achieved.

Phase 8 learning review — Data and KPIs

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Purpose and essential concept

Grain is what one row represents. Counting errors is only meaningful when the denominator is clear.

Practical task

Recalculate the four-order fixture manually and compare the output.

Three understanding questions

1. Why does joining deliveries and allocations multiply rows?

2. How do open overdue orders affect on-time delivery?

3. Why are synthetic patterns not business findings?

Status: answers not supplied; understanding not assessed. Review before claiming mastery. Building all phases was authorized; this does not substitute for your learning review.

Phase 9 learning review — UAT

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Purpose and essential concept

UAT checks business needs in a system with evidence. A design walkthrough checks the planned logic.

Practical task

Walk TC-014 and explain partial versus residual payment.

Three understanding questions

1. Why are all SAP cases Not executed?

2. What is the difference between severity and priority?

3. What evidence is needed to mark a test passed?

Status: answers not supplied; understanding not assessed. Review before claiming mastery. Building all phases was authorized; this does not substitute for your learning review.

UAT plan and honest validation status

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Purpose

User acceptance testing checks whether business users can meet agreed needs in the configured solution. A requirements walkthrough checks the design. Executing a spreadsheet or data script does not execute SAP UAT.

Entry criteria

Reviewed requirements/rules and traceability; known release/configuration and scope; non-production tenant; authorized test roles; controlled synthetic master/transaction data; reset approach; approved expected results; evidence storage and agreed severity. These entry conditions have not been met for SAP execution.

Exit criteria for a future SAP test cycle

All Must scenarios executed with evidence; no open critical/high issue preventing safe order, financial or access processing; medium/low residual issues explicitly accepted by accountable owners with workarounds; regression/coverage reviewed; actual business acceptance recorded. No fabricated approval or acceptance is supplied here.

Severity

Critical: unauthorized financial release, severe data corruption or widespread inability to operate safely. High: core flow blocked or materially wrong quantity/value without safe workaround. Medium: restricted scenario affected with a controlled workaround. Low: cosmetic/help issue without material processing impact. Priority is how soon to fix, based on business timing and impact; it is not automatically identical to severity.

Execution records

All 22 cases in uat-cases.csv remain **Not executed**. No SAP defects or screenshots exist. Synthetic data and KPI checks are separately recorded in phase-08/data/data_checks.csv. The fixture oracle exercises analytical logic related to TC-018 but does not change its SAP status. Document reviews are recorded in walkthrough.md as design issues.

For later execution, save actual inputs, time, user role, relevant document IDs, observed result and evidence path. If blocked before the action, mark Blocked with reason—not Passed. Do not use generated concept artwork as evidence.

Executed desk-based design walkthrough

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Execution type: AI-assisted document/data walkthrough, 11 October 2026. No SAP interaction and no stakeholder participation.

Walkthrough	Design observation	Resolution / status
TC-004/005: 100 ordered, 60 available	On-hand vs available wording can lead to unsupported promise	Base scenario stays unresolved; test variant explicitly assumes availability and permission
TC-011: dispatch Monday, posting Tuesday, receipt Wednesday	One delivery date would obscure service and posting lag	RULE-006 and dataset separate all three events
TC-014: £900 toward £1,500	Analytical remaining balance could be mistaken for SAP invoice clearing	SAP-006 distinction and net-only model limitation added
TC-018: multiple deliveries and payments	Raw child joins multiply order values	Aggregate to order/invoice grain; independent SQL and fixture checks executed
TC-020/021: audit and usability	Offline viewer cannot prove production permissions or user task success	System controls and actual user review remain pending

DI-001 (resolved design issue): date semantics needed separation. DI-002 (resolved design issue): payment model needed explicit distinction from SAP open-item postings. DI-003 (open design issue): full receipt-data capture and audit retention need tenant/policy validation. These are not executed SAP defects.

Illustrative business case — no achieved savings

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Decision question

What is the proportionate next step for the three process problems? A free portfolio does not imply a real ERP implementation is free. All implementation costs below are hypothetical business planning placeholders, distinct from the learner's £0 required software-purchase route.

A: Improve manual controls

One_off_GBP: 2000

Annual_running_GBP: 500

Minutes_saved_per_order: 2

Annual_orders_assumed: 6000

Loaded_hourly_rate_GBP: 25

Annual_capacity_hours: 200.0

Capacity_equivalent_GBP: 5000.0

Cash_savings_GBP: Unproven

Cost_basis: Illustrative planning placeholder, not supplier quote or SAP price

B: Standardise process and shared reporting

One_off_GBP: 8000

Annual_running_GBP: 1500

Minutes_saved_per_order: 5

Annual_orders_assumed: 6000

Loaded_hourly_rate_GBP: 25

Annual_capacity_hours: 500.0

Capacity_equivalent_GBP: 12500.0

Cash_savings_GBP: Unproven

Cost_basis: Illustrative planning placeholder, not supplier quote or SAP price

C: SAP-aligned future state

One_off_GBP: 60000

Annual_running_GBP: 18000

Minutes_saved_per_order: 8

Annual_orders_assumed: 6000

Loaded_hourly_rate_GBP: 25

Annual_capacity_hours: 800.0

Capacity_equivalent_GBP: 20000.0

Cash_savings_GBP: Unproven

Cost_basis: Illustrative planning placeholder, not supplier quote or SAP price

One-off categories include discovery, design, data cleansing, configuration or spreadsheet setup, integration, testing, training and contingency. Annual categories include administration, support, subscriptions if applicable, monitoring and regression. Option C is a broad allowance, not a SAP licence quote; commercial discovery would be mandatory. These estimates do not come from our synthetic dataset.

At 6,000 assumed annual orders, 5 minutes saved each equals 500 hours/year. At £25/hour, this is £12,500 of **capacity equivalent**, not cash savings. Staff can use time for other work; payroll falls only if a specific cost-removal decision exists. Do not also count the same minutes as fewer errors, faster billing or headcount savings. Earlier billing might affect working capital, but no cash-flow benefit is calculated without actual payment behaviour and financing assumptions.

Sensitivity and investment caution

For Option B, 2–8 minutes saved and £20–£30/hour gives 200–800 hours and £4,000–£24,000 capacity equivalent per year. At the central assumptions, dividing £8,000 by (£12,500−£1,500) would look attractive, but calling that cash payback would be misleading because salary spending has not been shown to fall. No cash ROI or payback is claimed. To calculate one meaningfully, establish incremental cash benefits, costs, timing and a defensible horizon first.

Recommendation

Start with Option A's ownership and reconciliation controls, then validate Option B against genuine baseline evidence. Keep the SAP-aligned design as a reference option requiring system fit, entitlements, costs and business readiness assessment. This sequence does not presume SAP is the best investment. Benefits depend on adoption, clean data, operating discipline and actual opportunities to redeploy capacity.

Risks, change and support

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Risk scoring uses illustrative 1–5 likelihood and impact scales; 1 is low, 5 high, score is their product. Scores guide discussion, not quantified loss forecasts. Ten risks are in risks.csv. RISK-007 and RISK-010 remain particularly important because desk research is not system evidence or learner mastery.

Sales

Change: Promise only checked quantity/date; retain original promise; correction queue

Training: 45-minute NS-ORD-1001 and missing-data exercise

Communication: Before trial: explain why checks protect promise quality

Support: Sales key user → process owner → IT if technical

Warehouse

Change: Separate physical dispatch from posting; resolve quantity mismatch

Training: 45-minute partial pick and delayed-posting exercise

Communication: Daily handoff checklist and unresolved stock review

Support: Warehouse key user → Warehouse owner → IT

Finance

Change: Own billable population; controlled correction and payment handling

Training: 60-minute billing retry, credit release and partial-payment cases

Communication: Daily unbilled/credit/payment review; weekly control issues

Support: Finance key user → Finance owner → IT

Customer Service

Change: Use authoritative status; capture partial agreement and next action

Training: 30-minute status enquiry and customer agreement role-play

Communication: Short quick guide; weekly feedback session in pilot

Support: Customer Service key user → relevant process owner

IT

Change: Support role matrix, data contracts, refresh and evidence retention

Training: 60-minute access, failed-refresh and reconciliation exercise

Communication: Release notes and change-impact review before changes

Support: IT support → authorized platform support where applicable

Training durations are proposed session lengths based on a few concrete scenarios, not measured learning requirements. Use job aids, a practice task, observation and feedback; attendance alone does not establish competence. Communications should explain the changed rule, why it matters, who owns the next step and where to get help. No training has been delivered.

Suggested support: role key user triages process questions; business owner decides policy; IT handles access/integration/report failures; specialist/vendor support is a possible later dependency, not a purchased service. Critical access/financial integrity problems require immediate containment and Finance/IT escalation; other issues enter a daily triage queue. Actual service levels and support coverage require agreement.

Phase 10 learning review — Risks, change and business case

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Purpose and essential concept

A business case compares options, costs and uncertain benefits. Saved minutes are capacity unless spending actually falls.

Practical task

Change the illustrative minutes saved and inspect sensitivity.

Three understanding questions

1. Can staff capacity be counted again as cash savings?

2. Why might better manual controls be the first investment?

3. What would make payback meaningful?

Status: answers not supplied; understanding not assessed. Review before claiming mastery. Building all phases was authorized; this does not substitute for your learning review.

Case study and lessons learned

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Context

Independent simulated analysis of NorthStar Manufacturing UK, a one-company, one-warehouse domestic stock seller. Supplied problems: poor visibility, stock delays and manual billing handoffs. The learner's quality-review and error-analysis background motivates attention to controls and evidence; no real client engagement is claimed.

Analysis and decisions

The pack uses explicitly invented discovery to explore conflicting promises, partial deliveries, credit and billing ownership. AS-IS and TO-BE flowcharts distinguish physical events from postings. Three business requirements decompose into 15 functional and five non-functional requirements, 15 stories and 22 planned UAT cases. Traceability links problems, process changes, tests and measures. Official SAP references support broad standard-process fit; exact configuration and several controls remain unresolved.

The main decisions were to preserve the original promise date, require partial-delivery agreement, assign Finance the billable-population review, distinguish partial payment from clearing, and aggregate child records before combining measures. Standard processes and manual controls were considered before extensions.

Evidence produced

The reproducible dataset contains 300 synthetic orders, 285 shipments, 262 invoices and 197 allocations. Six KPI calculations agree between Python and SQLite; 24 data checks passed. The offline viewer exposes results and records but is not an SAP application. All 22 SAP UAT cases remain Not executed.

Limitations and next steps

No real stakeholder research, tenant demonstration, implementation or measured savings. Data is simplified to one line per order and net-only finances; it omits corrections, VAT and returns. A real project needs agreed policies, broader representative data, configuration/integration/security testing and business acceptance. Learner understanding and personal contribution still need review before resume use.

Lessons from building the pack

An attractive dashboard is not evidence of a sound denominator. Posted goods issue is not proof of receipt. A plausible root cause is still a hypothesis. A linked analytical payment amount is not the same as an SAP clearing document. A generated test list is not executed UAT. These are artifact-development observations, not retrospective lessons from a live implementation.

Executive presentation — ten slides

Independent simulated portfolio; editable PPTX in downloads.

1. NorthStar Manufacturing UK

SAP S/4HANA Order-to-Cash

Business Process Analysis & Transformation

- Independent simulated portfolio

- Public Edition 2602 research reference

- Free documentation and synthetic-data route

Speaker note: Explain that no real client, production configuration or implementation is represented.

2. The business problem

Three connected operational problems

- Poor visibility → repeated enquiries and unclear ownership

- Stock delays → unsupported promises and partial-delivery decisions

- Manual billing handoffs → eligible deliveries waiting for review

Speaker note: Consequences are hypotheses from the supplied fictional scenario, not real observed findings.

3. Scope and representative order

One company · one warehouse · domestic B2B · GBP

- NS-ORD-1001: 100 brackets × £25 = £2,500 net

- 60 physically on hand; availability remains unresolved

- Requested receipt 9 October; no confirmed promise invented

- Order receipt through payment allocation and reporting

Speaker note: Manufacturing planning, procurement redesign, international trade, complex tax, full migration and deployment are excluded.

4. Discovery and process choices

Role-play reveals decisions, not stakeholder approval

- Validate data before confirmation; retain the original promise

- Require agreement for partial delivery; keep remainder open

- Finance owns credit release and billable-population review

- Separate physical dispatch, goods-issue posting and receipt

Speaker note: Explain alternative causes for late billing; email handoff is a hypothesis, not a proven root cause.

5. Requirements to validation

One traceable evidence chain

- 23 requirements: 3 business + 15 functional + 5 non-functional

- 15 stories and 12 rules with explicit exceptions

- Example: PROB-002 → BR-002 → FR-004 → PC-004

- US-004 → TC-005/006 → KPI-003

Speaker note: All requirements and stories are proposed. Trace links exist but actual business validation is pending.

6. SAP standard-first assessment

Documented capability is not demonstrated configuration

- Seven official sources reviewed; two pages explicitly unverified

- BD9 stock-sales reference and payment handling researched

- 15 functional fit assessments remain desk-based

- Validate exact roles, corrections, receipt capture and audit coverage

Speaker note: Reference is Public Edition 2602; Learning pages are not release-pinned. Entitlements and configuration are not established.

7. Synthetic baseline

Reproducible analysis, not achieved performance

Speaker note: Single-line orders, net-only finance, seeded data. On-time denominator includes overdue open orders. Cycle time excludes unfinished orders; billing mean excludes unbilled deliveries.

8. Validation and remaining gates

Honest execution status

- 24 synthetic data checks passed; six Python/SQL KPIs agree

- 22 SAP UAT cases designed; 0 executed

- Desk walkthrough issues kept separate from SAP defects

- Learner explanation, real discovery and tenant testing remain pending

Speaker note: Do not turn data check success into a passed SAP UAT case or personal experience claim.

9. Options and illustrative economics

Improve controls before assuming a large investment

- Compare manual controls, process standardisation and SAP-aligned design

- 6,000 orders × 5 minutes = 500 hours annual capacity

- At £25/hour: £12,500 capacity equivalent, not cash savings

- No cash ROI/payback claimed; costs are planning placeholders

Speaker note: Recommend manual ownership/reconciliation first, then validate process standardisation. No SAP price quote or achieved benefit is represented.

10. What this portfolio demonstrates

A reviewable foundation, with explicit limits

- Problem → evidence → requirements → design → validation reasoning

- Reproducible synthetic analysis and source-labelled SAP research

- Next: learner review, policy validation and system demonstration

- AI drafted/generated artifacts; personal contribution must be stated accurately

Speaker note: Final learning gate remains open until the learner can explain decisions without reading a script. No employment guarantee.

Phase 11 learning review — Packaging

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Purpose and essential concept

A portfolio tells a defensible evidence story, not a list of files.

Practical task

Give a two-minute tour linking a problem to a requirement, story and test.

Three understanding questions

1. What is the strongest evidence in this pack?

2. Which limitation should an interviewer hear?

3. What would you remove to reduce unnecessary documentation?

Status: answers not supplied; understanding not assessed. Review before claiming mastery. Building all phases was authorized; this does not substitute for your learning review.

Truthful resume and interview preparation

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Contribution check — required before using personal claims

The artifacts and automated analyses were produced with AI assistance. The learner has not yet supplied review answers or demonstrated independent reproduction. Therefore the statements below are **project-description drafts**, not approved personal achievement claims. Use “I analysed/designed/tested” only after personally performing and explaining the relevant work.

Four evidence-supported project-description bullets

- Independent simulated SAP S/4HANA Order-to-Cash portfolio covering a fictional UK manufacturer, with current/future process flows and a source-labelled SAP reference assessment.

- Portfolio includes 23 requirements, 15 user stories and 22 planned UAT cases linked through traceability; SAP test cases are labelled Not executed.

- Reproducible analysis covers 300 synthetic orders and six KPIs, with separate order, delivery, invoice and payment-allocation grains and Python/SQLite reconciliation.

- Compares manual-control, process-standardisation and SAP-aligned options using illustrative costs and capacity assumptions, without claiming achieved savings.

60-second overview to adapt truthfully

“My portfolio is an independent simulation of Order-to-Cash at a fictional UK manufacturer. It focuses on order visibility, stock delays and billing handoffs. The pack connects discovery assumptions to process changes, requirements, stories and planned tests. It uses official SAP Public Edition sources as a reference, not a live implementation. A 300-order synthetic dataset demonstrates six KPIs and the importance of avoiding duplicate values across shipments and payments. AI helped draft and generate the artifacts; my personal contribution so far is [state only work actually completed]. I am working through the decisions and checks so I can explain them independently. All SAP UAT cases remain unexecuted.”

Five-minute walkthrough

0:00–0:45: scenario, supplied problems and evidence boundary. 0:45–1:30: NS-ORD-1001 and on-hand versus available stock. 1:30–2:15: AS-IS hypothesis and TO-BE ownership. 2:15–3:00: FR-004 → US-004 → TC-005/006 → KPI-003. 3:00–4:00: dataset grains, original-promise denominator and executed checks. 4:00–4:40: desk-based SAP fit, unexecuted UAT and option comparison. 4:40–5:00: personal contribution, limitations and next validation step.

Interview questions and challenges

1. BA: How would you turn “poor visibility” into a requirement? Follow-up: which role, fields and freshness matter? Strong answer names observable outcomes and evidence.

2. Discovery: Sales wants Friday delivery, Warehouse cannot confirm. What do you do? Follow-up: how do you distinguish request, estimate and commitment?

3. Process: What proves the customer received the goods? Follow-up: what if dispatch and posting occur on different dates?

4. SAP: Why choose Public Edition as a reference? Follow-up: which exact claims still require tenant verification?

5. Controls: Who can release credit and why? Follow-up: what if a separate order block remains?

6. Requirements: Give a negative acceptance criterion for partial delivery. Follow-up: what happens to the remaining quantity?

7. Data: How can a join inflate order value? Follow-up: show the grain before and after aggregation.

8. KPI: Should an open overdue order be excluded from on-time delivery? Follow-up: why retain the original promise date?

9. UAT: Can TC-018 be Passed because Python checks passed? Follow-up: what evidence would a system test require?

10. Finance: Does a partial payment clear the original invoice? Follow-up: contrast partial and residual treatment using SAP-006.

11. Business case: Are 500 saved hours £12,500 cash savings? Follow-up: what decision would make cash benefits real?

12. Integrity: What did you do and what did AI do? Follow-up: reproduce one calculation without relying on the generated explanation.

Coaching rubric

For each answer record 0 = incorrect/no explanation, 1 = partial, 2 = correct with example and limitation. Do not assign scores before answers exist. Re-teach stock availability, event dates, grain or test status if confused. Readiness requires explaining the main decisions without a script, not memorizing every file. Employment is not guaranteed.

Phase 12 learning review — Interview preparation

Independent simulated portfolio · 2026-10-11 · AI-assisted draft; learner review pending.

Purpose and essential concept

A credible answer explains your decisions and actual contribution. AI drafting is not personal execution.

Practical task

Record a 60-second overview without reading the script.

Three understanding questions

1. Which work did you personally validate?

2. Which claim would overstate SAP experience?

3. How would you investigate a result you disagree with?

Status: answers not supplied; understanding not assessed. Review before claiming mastery. Building all phases was authorized; this does not substitute for your learning review.

Assumptions, constraints and dependencies

Updated 8 October 2026. Supplied scenario facts describe a fictional business, not verified real operations.

ID	Type	Entry	Status / treatment
ASM-001	Supplied scenario fact	One company, one warehouse, domestic stock sales, GBP reporting	Fixed initial scope
ASM-002	Fictional assumption	Current work uses fragmented records and manual communications	Specify during current-state analysis
ASM-003	Learning assumption	4–6 study hours per week	Await learner response
ASM-004	Fictional assumption	Partial delivery requires defined customer agreement or policy	Policy unresolved
ASM-005	Fictional assumption	NS-ORD-1001 values in representative-order.md	Stable base scenario; changes must be logged
CON-001	Constraint	SAP access unknown	Documentation/data route remains available
CON-002	Constraint	No real client records, interviews, approvals or production evidence supplied	Label all simulation
CON-003	User constraint	All project work must avoid required paid tools or services	Use free documentation/data route; no paid SAP access, Office licence or hosting required
DEP-001	Dependency	Stock availability depends on replenishment and record accuracy	Record impact; procurement redesign excluded
DEP-002	Dependency	Billing/allocation require customer, pricing, tax and payment information	Define data and controls later
DEP-003	Dependency	SAP behaviour varies by edition, release, configuration and integration	Verify claims before fit assessment

No numerical business performance target or achieved improvement has been established.

Full-build additions — 11 October 2026

ID	Type	Entry	Status / treatment
ASM-006	Fictional operating policy	Daily Finance queue review; daily management snapshot before 09:00 Europe/London	Proposed business-local cadence; not a service-level claim
ASM-007	Analytical assumption	One line per order, GBP integer pence, all finance net-only with VAT omitted	Explicit teaching limitation
ASM-008	Analytical assumption	Synthetic snapshot 11 October 2026; dates are UK business-local calendar dates	Actual future events omitted; promises/due dates may be future
ASM-009	Business-case assumption	6,000 annual orders and £25 loaded hourly cost	Illustrative; independent of synthetic baseline
ASM-010	Learning assumption	Broad junior BA emphasis with ERP/process skills retained	Primary target role still unresolved
CON-004	Evidence constraint	Learner has not personally reproduced or explained artifacts in this conversation	Conditional resume drafts; no competence claim
DEP-004	Integration dependency	Receipt evidence, audit coverage and retention require validation	No automatic carrier integration or retention guarantee

registers/decisions.md

Decision log

ID	Date	Decision	Rationale / status
DEC-001	2026-10-08	One phase at a time	User-directed; review understanding before advancing
DEC-002	2026-10-08	Documentation and synthetic analysis if SAP access unavailable	Keeps work feasible; access still unknown
DEC-003	2026-10-08	Provisionally reference SAP S/4HANA Cloud Public Edition	Defined research boundary; reference release unresolved; revisit against access
DEC-004	2026-10-08	Assess standard processes before extensions	Avoid unnecessary complexity; no capability fit yet verified
DEC-005	2026-10-08	Defer numeric improvement targets	Need measurement definitions and baseline first
DEC-006	2026-10-08	Store editable Phase 1 files in cloud workspace	User asked to work on cloud; does not establish SAP access or publication approval
DEC-007	2026-10-08	Select documentation and synthetic-data route with no required paid software	User requires free work; use Markdown, browser documents, LibreOffice-compatible copies, CSV and later Python/SQLite; SAP access remains unknown
DEC-008	2026-10-11	Build all twelve phases now	User explicitly requested the whole project; learning reviews remain pending
DEC-009	2026-10-11	Pin SAP reference to Public Edition 2602	Official feature-scope PDF reviewed; learning sources are unpinned; no tenant access inferred
DEC-010	2026-10-11	Use a 300-order single-line, net-only analytical dataset	Keeps reproducibility manageable; not a full UK tax/receivables ledger
DEC-011	2026-10-11	Provide an offline portfolio viewer, not hosted production software	Preserves free-tool constraint and original instruction not to publish without request
DEC-012	2026-10-11	Keep representative order unresolved in baseline	Design/test variants are separate; do not invent actual fulfilment
DEC-013	2026-10-11	Draft resume statements conditionally	Artifacts generated with AI; personal validation and learning review not established

ID	Date	Decision	Rationale / status
DEC-014	2026-10-11	Add real-browser screenshots, explanatory PDFs and GitHub-ready root entry	User requested local run evidence and a repository they can push; no remote publication performed

Working glossary

Term	Meaning
Order-to-Cash	Activities from receiving a customer order through collecting and matching payment
Requirement	Clear statement of a needed outcome or behaviour
Business rule	Policy or condition governing a business decision or action
Control	Check or restriction that prevents or detects an error
Exception	Situation requiring handling outside the normal flow
Root-cause hypothesis	Possible underlying explanation still requiring evidence
Acceptance criterion	Observable condition for judging whether a requirement is met
KPI	Key performance indicator: defined measure used to monitor performance
UAT	User acceptance testing: checking whether a solution meets business needs
Traceability	Links connecting problems, requirements, design and validation
ERP	Enterprise resource planning: software supporting connected business activities and records
BRD	Business requirements document: explanation of business needs, scope and requirements
AS-IS / TO-BE	Current process / proposed future process
Backlog	Prioritised collection of work or product needs
User story	Short statement of a user's need and its business value
Goods-issue posting	System record of an outbound stock movement; exact SAP effects require verification
Payment allocation	Matching received money to the correct invoice or account
Synthetic data	Artificial records generated for analysis or testing
Edition / release	Product offering and operating model / version of that offering
Fit-to-standard	Comparing business needs with standard system/process capabilities

These are learning definitions, not a verified SAP capability register.

Terms introduced in the full build

Term	Meaning
Grain	What one data row represents
Primary / foreign key	Unique row identifier / reference to another table's row
Fan-out	Multiplication of parent rows when several child tables are joined
Business partner	Common identity and role-specific master information; see SAP-005
ATP	Available to promise: date/quantity confirmation concept; see SAP-002
PGI	Post goods issue; recording an outbound stock movement, not customer receipt
Receivable	Amount owed by a customer
Partial payment	Payment linked to an invoice while original invoice remains open; see SAP-006
Residual item	Remaining balance item after clearing the original invoice; see SAP-006
On-account payment	Payment recorded without matching a particular invoice
RACI	Responsible, Accountable, Consulted and Informed responsibilities
Segregation of duties	Separating conflicting actions, such as requesting and approving release
Capacity released	Staff time available for other work; not necessarily cash saved
Severity / priority	Impact of a defect / urgency of resolving it
Configuration / extension	Setting standard behaviour / adding capability beyond the chosen standard

Current progress — 11 October 2026

Independent simulated portfolio. Full artifact build authorized by the user. Deliverable creation does not establish learner mastery, stakeholder approval or production readiness.

Phase	Artifact status	Validation / limitation
1 Charter and plan	Built; original charter preserved with supersession note	SAP access, actual study hours and target role remain unknown
2 Discovery	Built	Explicit role-play; no people interviewed
3 AS-IS	Built	Hypotheses and fictional tools; process flowchart, not BPMN
4 Requirements	Built	23 requirements, 12 rules; no stakeholder approval
5 SAP / TO-BE	Built	Public Edition 2602 reference; seven reviewed sources and two unverified pages
6 Fit-gap	Built	Desk-based; no tenant demonstration
7 Backlog	Built	15 proposed stories; not delivered SAP functionality
8 Data and KPIs	Built and executed	300 synthetic orders; six Python/SQL KPIs; data-check evidence available
9 UAT	Design built	22 SAP cases Not executed; separate desk walkthrough/data checks
10 Risks/change/case	Built	Ten open risks; illustrative cost/capacity assumptions
11 Packaging	Built	Offline viewer, editable files, presentation and case study; not published
12 Resume/interview	Conditional drafts built	Actual learner contribution and oral review still pending

Outstanding quality gates

- Learner answers and independent explanation have not been reviewed.

- SAP access is unknown; no paid access required or purchased.

- No real discovery, policy agreement, configured system tests or business acceptance.

- Data is single-line, net-only and synthetic; receipt integration and exact audit coverage unresolved.

- Real local Chromium checks and actual screenshots are recorded in quality/browser-checks.json; LibreOffice visual rendering remains unverified. The browser record identifies the exact tested HTML hash.

Artifact preparation is complete subject to the recorded checks and limitations. The project must not be labelled fully complete under the original final gate until learner understanding is reviewed. It is not production ready.

Handover update — 12 October 2026

Actual local Chromium checks and nine UI screenshots completed. Root index.html, local-server script, GitHub Pages guide and PDF handover added. No remote publishing occurred. Exact test timing and current source hashes are in quality/browser-checks.json.

Final quality gate — 11 October 2026

Independent simulated portfolio. **Artifacts prepared; original project completion gate remains open. Not production ready.**

Executed evidence

24 Python/SQLite synthetic-data checks passed. Six KPI calculations agree before rounding. Sixteen catalogue consistency checks below passed. These are data/document checks, not SAP UAT. Office files are reopened and the offline viewer receives JavaScript logic smoke checks in the packaging step; see artifact-checks.json. Actual local Chromium rendering and 20 UI checks are recorded in browser-checks.json, with screenshots and the tested HTML hash. Real role permissions and LibreOffice visual checks were not performed.

ID	Check	Result	Evidence
QC-001	Catalogue unique IDs	Passed	3 BR + 15 FR + 5 NFR; 15 stories; 22 test designs
QC-002	All three business problems traced	Passed	Every supplied problem has a trace row
QC-003	Every functional requirement traced	Passed	No functional requirement missing from trace
QC-004	Story/test references valid	Passed	All story requirement and test references resolve
QC-005	Trace and test bidirectional	Passed	Trace test links include the same requirement
QC-006	All tests link valid requirements	Passed	No dangling test requirements
QC-007	All NFRs covered	Passed	All five NFRs linked to planned cases
QC-008	SAP execution labels honest	Passed	No invented test result or SAP defect
QC-009	Seven exceptions covered	Passed	Includes positive and negative partial-delivery permission; pre/post dispatch cancellation
QC-010	Rules and process change IDs	Passed	All 12 rules referenced; 15 unique changes
QC-011	Fit assessment covers FRs	Passed	15 desk-based fit assessments; no claimed tenant demo
QC-012	Source evidence completeness	Passed	Official URLs/access date and demonstration boundary recorded
QC-013	Unverified sources explicit	Passed	Two unreadable Help pages not treated as verified capability

ID	Check	Result	Evidence
QC-014	Synthetic data checks passed	Passed	24 checks incl. independent Python/SQL and hand-calculated fixture
QC-015	Required files present	Passed	Core narratives/maps exist
QC-016	Learning questions retained	Passed	All twelve learning reviews present; none marked passed

Manual desk review

Requirements address the three supplied problems; core exception paths appear in both flowcharts and planned tests. PC changes map to the named requirements and proposed stories. Seven exception categories have test coverage. Standard-process claims cite the SAP evidence register; two unreadable pages remain unverified. Business targets have not been achieved. Financial amounts are explicitly net-only, and capacity assumptions are not cash savings.

Remaining weaknesses

1. Learner answers, personal contribution and independent explanation have not been reviewed. Conditional resume descriptions cannot yet be used as personal execution claims.

2. Discovery and policies are invented role-play. No stakeholders have validated priorities, ownership, thresholds or acceptance.

3. SAP is a documentation reference: no tenant, configuration, authorization, integration or UAT evidence. All 22 SAP cases remain Not executed.

4. Synthetic data is single-line and net-only; VAT, corrections, returns and the full SAP journal/open-item model are absent. These need broader modelling for a real project.

5. Receipt capture, audit retention and exact billing/credit controls remain unresolved. Learning pages are not release-pinned even though the reference PDF is 2602.

6. Viewer is read-only learning software with embedded synthetic data. It does not enforce production authorization or implement the transactional requirements. Actual local browser checks are recorded separately; Office visual QA, other-browser coverage and comprehensive accessibility/security review remain outstanding.

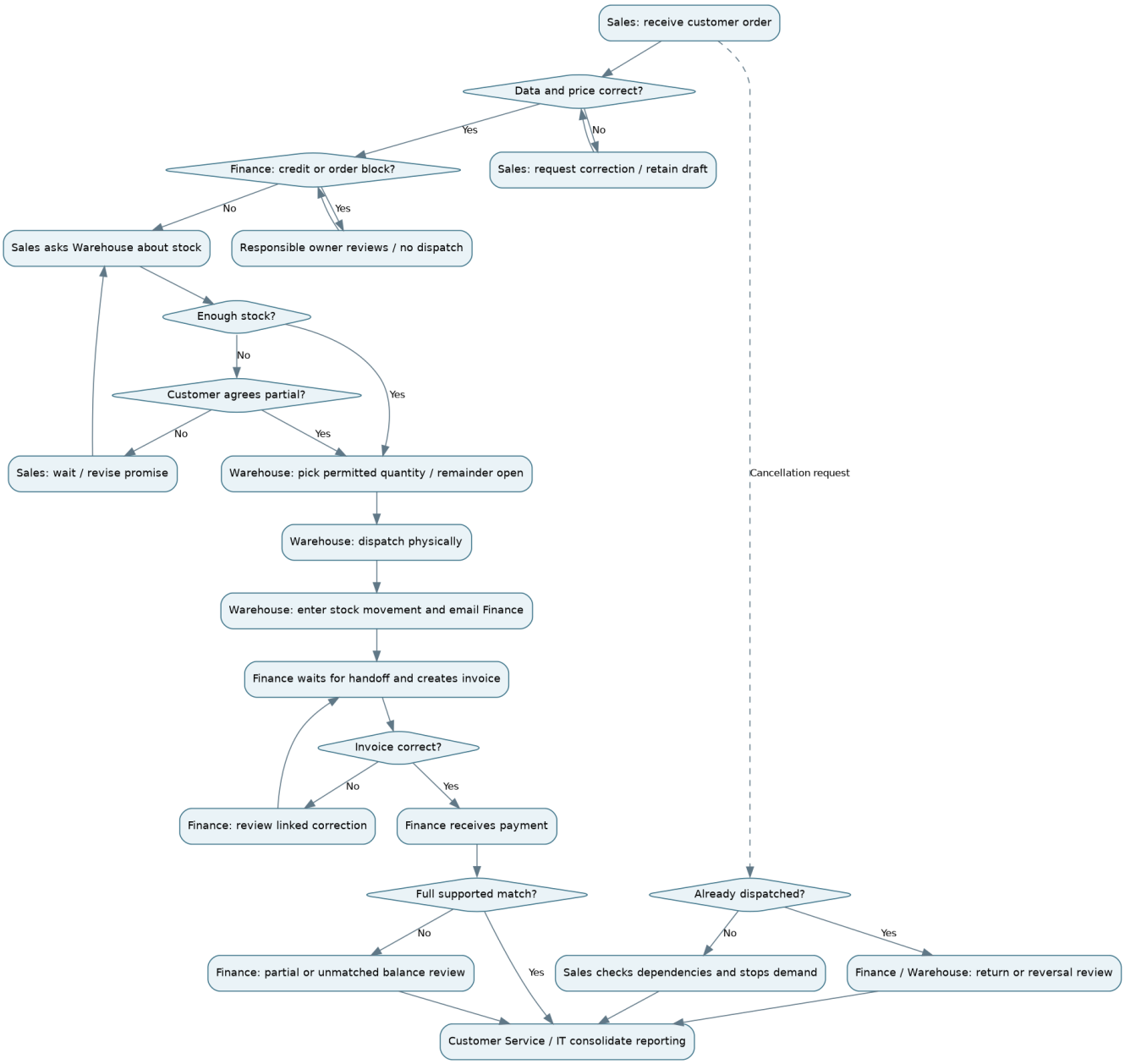
7. Business-case volumes, handling minutes and costs are illustrative. No cash ROI, achieved benefit or investment approval is claimed.

Honest completion statement

The requested twelve-phase artifact pack is built. Production readiness is not established, and full portfolio completion under the user's original gate requires the learner to explain key decisions without a script. Use the phase learning reviews next, then revise personal claims to match actual work completed.

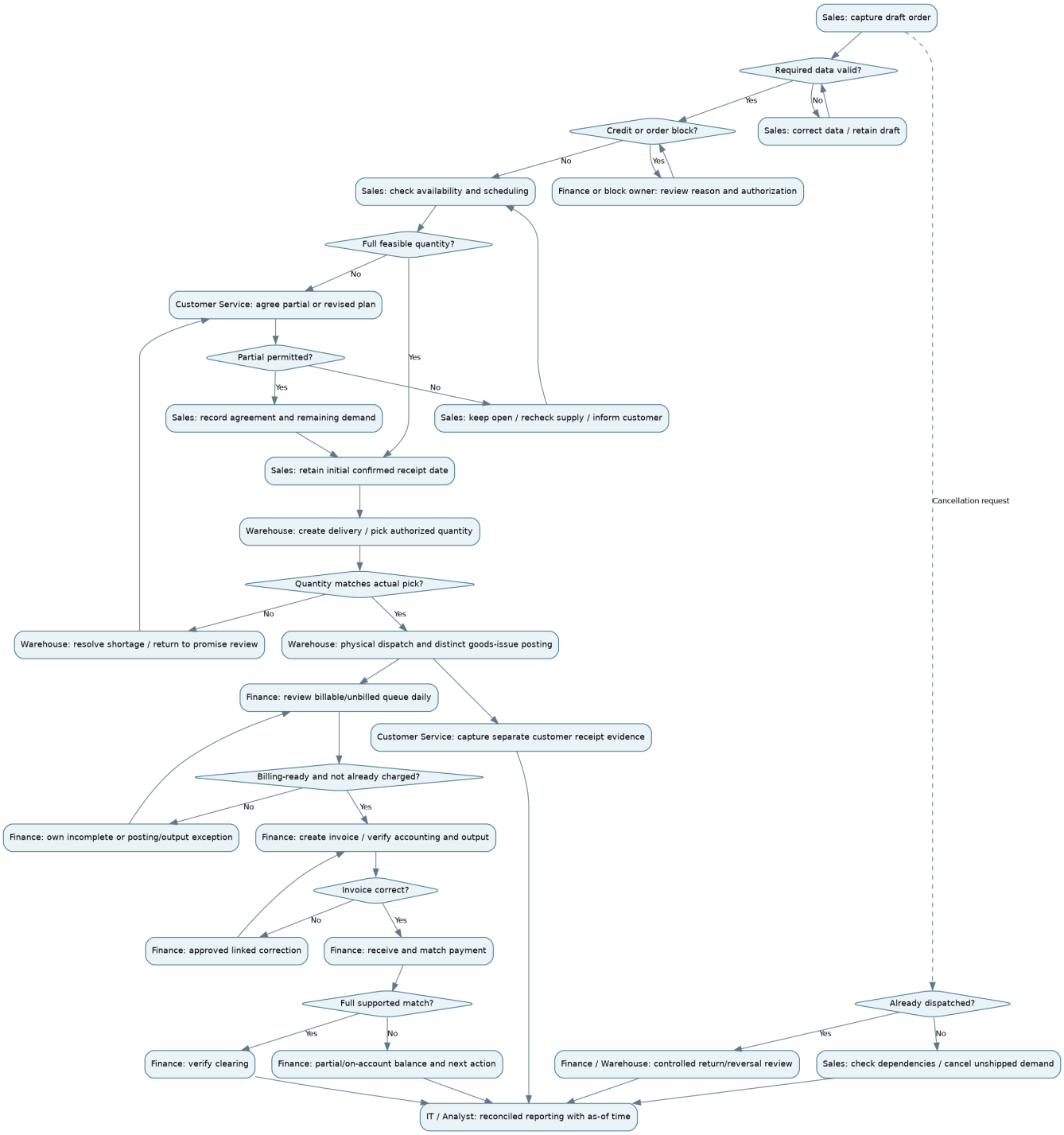
AS-IS process flowchart

Entire diagram for orientation. Use the supplied SVG for readable zoom. This is a flowchart, not BPMN.



TO-BE process flowchart

Entire diagram for orientation. Use the supplied SVG for readable zoom. This is a flowchart, not BPMN.



Appendix — Business requirements and acceptance

Editable source: phase-04/requirements.csv

1. BR-001

ID: BR-001

Type: Business

Statement: Enable accountable, traceable order and payment status from intake through allocation and reporting.

Rationale: Reduce uncertainty and prevent lost exceptions.

Source: PROB-001; SIM-CS-01; SIM-FIN-04

Priority: Must

Priority reason: Core visibility problem

Acceptance: Representative normal, blocked, cancelled and partially paid cases expose current stage, owner, remaining quantity/balance and evidence gaps.

Dependencies: Master data; linked documents; agreed role access

Validation: Trace review and planned UAT TC-001;TC-014;TC-017

2. BR-002

ID: BR-002

Type: Business

Statement: Base fulfilment commitments and release on feasible supply, customer agreement and financial controls.

Rationale: Avoid unsupported promises and unauthorized shipments.

Source: PROB-002; SIM-WH-01; SIM-FIN-01

Priority: Must

Priority reason: Core fulfilment/control problem

Acceptance: Stock shortage, partial permission and block cases follow rules; measures KPI-001/002/003 use the agreed definitions; no achieved improvement claimed.

Dependencies: Stock accuracy; credit policy; transit assumptions

Validation: Planned UAT TC-004 through TC-008; synthetic KPI checks

3. BR-003

ID: BR-003

Type: Business

Statement: Make eligible fulfilled deliveries visible to Finance for timely, accurate and traceable billing.

Rationale: Remove avoidable handoff waiting without billing incorrectly.

Source: PROB-003; SIM-FIN-02; SIM-FIN-03

Priority: Must

Priority reason: Core billing problem

Acceptance: Eligible delivery appears in review population without an email; correction/duplicate/error paths retain links and owner; KPI-004 includes unbilled context.

Dependencies: Posting accuracy; billing relevance; Finance ownership

Validation: Planned UAT TC-012;TC-013;TC-022

4. FR-001

ID: FR-001

Type: Functional

Statement: The proposed process/solution shall prevent order confirmation until required customer and order fields are valid.

Rationale: Customer data validation addresses PROB-001 and supports BR-001.

Source: SIM-SALES-01 (role-play)

Priority: Must

Priority reason: Control or essential end-to-end capability

Acceptance: Missing ship-to address or inactive customer prevents confirmation, identifies the field and preserves the draft; corrected data can be rechecked

Dependencies: Customer owner; approved field list; RULE-001

Validation: Design walkthrough; planned SAP tests TC-001;TC-002;TC-003

5. FR-002

ID: FR-002

Type: Functional

Statement: The proposed process/solution shall see quantities, dates, blocks, related documents and accountable owner in one order view.

Rationale: Order status visibility addresses PROB-001 and supports BR-001.

Source: SIM-CS-01 (role-play)

Priority: Must

Priority reason: Control or essential end-to-end capability

Acceptance: NS-ORD-1001 shows 100 ordered, availability unresolved, requested receipt 9 October and no invented confirmation; unknown facts are labelled

Dependencies: Document relationships; FR-003; NFR-003; RULE-002

Validation: Design walkthrough; planned SAP tests TC-017;TC-021

6. FR-003

ID: FR-003

Type: Functional

Statement: The proposed process/solution shall check usable supply and commitments before promising quantity and date.

Rationale: Availability-based promises addresses PROB-002 and supports BR-002.

Source: SIM-WH-01 (role-play)

Priority: Must

Priority reason: Control or essential end-to-end capability

Acceptance: 100 ordered with only 60 confirmed available cannot be promised as 100 on the requested date; confirmed quantity/date and check time are retained

Dependencies: Stock accuracy; scheduling lead times; RULE-003

Validation: Design walkthrough; planned SAP tests TC-001;TC-004

7. FR-004

ID: FR-004

Type: Functional

Statement: The proposed process/solution shall record customer agreement before splitting fulfilment.

Rationale: Partial delivery agreement addresses PROB-002 and supports BR-002.

Source: SIM-CS-02 (role-play)

Priority: Must

Priority reason: Control or essential end-to-end capability

Acceptance: With no partial-delivery permission, a 60/100 shipment is stopped for review; with recorded agreement, 60 can proceed and 40 remains open

Dependencies: FR-003; customer delivery policy; RULE-004

Validation: Design walkthrough; planned SAP tests TC-005;TC-006

8. FR-005

ID: FR-005

Type: Functional

Statement: The proposed process/solution shall hold blocked orders until the appropriate authorized review.

Rationale: Credit and order blocks addresses PROB-002 and supports BR-002.

Source: SIM-FIN-01 (role-play)

Priority: Must

Priority reason: Control or essential end-to-end capability

Acceptance: An unresolved credit or manual order block prevents dispatch release; Sales cannot release a credit block; Finance review records decision, reason and actor

Dependencies: Credit limits; NFR-001; precise block configuration; RULE-005

Validation: Design walkthrough; planned SAP tests TC-007;TC-008

9. FR-006

ID: FR-006

Type: Functional

Statement: The proposed process/solution shall record actual pick, dispatch, goods-issue posting and customer receipt separately.

Rationale: Fulfilment and posting evidence addresses PROB-002 and supports BR-002.

Source: SIM-WH-02 (role-play)

Priority: Must

Priority reason: Control or essential end-to-end capability

Acceptance: Picked/shipped quantities do not exceed authorized remaining quantity; dispatch and posting dates remain distinct; receipt is never inferred from posting

Dependencies: FR-004; delivery documents; receipt evidence; RULE-006

Validation: Design walkthrough; planned SAP tests TC-001;TC-005;TC-011

10. FR-007

ID: FR-007

Type: Functional

Statement: The proposed process/solution shall cancel unshipped demand without erasing transaction history.

Rationale: Controlled cancellation addresses PROB-001 and supports BR-001.

Source: SIM-SALES-02 (role-play)

Priority: Must

Priority reason: Control or essential end-to-end capability

Acceptance: Pre-dispatch cancellation records reason and removes demand from release queue; post-dispatch request routes to Finance/Warehouse review without deleting posted records

Dependencies: Document status checks; returns/reversal dependency; RULE-007

Validation: Design walkthrough; planned SAP tests TC-009;TC-010

11. FR-008

ID: FR-008

Type: Functional

Statement: The proposed process/solution shall review eligible dispatched deliveries without waiting for emailed paperwork.

Rationale: Billing readiness queue addresses PROB-003 and supports BR-003.

Source: SIM-FIN-02 (role-play)

Priority: Must

Priority reason: Control or essential end-to-end capability

Acceptance: A goods-issued delivery enters the billing review population; blocked/incomplete entries show reason and owner; unbilled age remains visible

Dependencies: FR-006; billing relevance; daily review calendar; RULE-008

Validation: Design walkthrough; planned SAP tests TC-001;TC-012;TC-022

12. FR-009

ID: FR-009

Type: Functional

Statement: The proposed process/solution shall bill eligible delivered quantities once and track posting/output status.

Rationale: Invoice creation and duplicate control addresses PROB-003 and supports BR-003.

Source: SIM-FIN-02 (role-play)

Priority: Must

Priority reason: Control or essential end-to-end capability

Acceptance: For 60 brackets at £25, net billing is £1,500; retry must not create a second active charge for the same quantity; posting/output failures remain visible

Dependencies: FR-008; pricing/tax master data; document linkage; RULE-009

Validation: Design walkthrough; planned SAP tests TC-001;TC-012;TC-013;TC-022

13. FR-010

ID: FR-010

Type: Functional

Statement: The proposed process/solution shall correct invoice errors through linked reviewed adjustments.

Rationale: Invoice correction addresses PROB-003 and supports BR-003.

Source: SIM-FIN-03 (role-play)

Priority: Must

Priority reason: Control or essential end-to-end capability

Acceptance: A wrong-price invoice is marked disputed and routed for review; correction preserves original reference and reason; no silent overwrite of a posted invoice

Dependencies: Correction method and tax treatment unresolved; RULE-009

Validation: Design walkthrough; planned SAP tests TC-013

14. FR-011

ID: FR-011

Type: Functional

Statement: The proposed process/solution shall match full and partial receipts while exposing unallocated balances.

Rationale: Payment allocation addresses PROB-001 and supports BR-001.

Source: SIM-FIN-04 (role-play)

Priority: Must

Priority reason: Control or essential end-to-end capability

Acceptance: Against £1,500 net-only illustrative invoice, £900 allocation leaves £600 analytical balance; unmatched £200 stays unallocated; allocation cannot exceed receipt or invoice balance

Dependencies: Customer reference; chosen partial vs residual policy; RULE-010

Validation: Design walkthrough; planned SAP tests TC-001;TC-014;TC-015;TC-016

15. FR-012

ID: FR-012

Type: Functional

Statement: The proposed process/solution shall assign each open exception to an owner and next action.

Rationale: Exception ownership addresses PROB-001 and supports BR-001.

Source: SIM-CS-01 (role-play)

Priority: Should

Priority reason: Supports operation; interim documented control possible

Acceptance: Each open exception has order reference, category, accountable role and review date; overdue reviews appear in queue without exposing unauthorized finance detail

Dependencies: Role matrix; escalation policy; RULE-002

Validation: Design walkthrough; planned SAP tests TC-017

16. FR-013

ID: FR-013

Type: Functional

Statement: The proposed process/solution shall calculate consistent measures without multiplying order values.

Rationale: KPI definitions and reconciliation addresses PROB-001 and supports BR-001.

Source: SIM-IT-01 (role-play)

Priority: Must

Priority reason: Control or essential end-to-end capability

Acceptance: Known fixture with two deliveries and two allocations retains one order value; cancelled/open/partial cases follow KPI contract; totals reconcile to source grains

Dependencies: Approved definitions; receipt capture; FR-006; FR-011; RULE-011

Validation: Design walkthrough; planned SAP tests TC-018

17. FR-014

ID: FR-014

Type: Functional

Statement: The proposed process/solution shall restrict sensitive actions to accountable roles.

Rationale: Role-based access addresses PROB-001 and supports BR-001.

Source: SIM-FIN-01 (role-play)

Priority: Must

Priority reason: Control or essential end-to-end capability

Acceptance: Warehouse cannot change credit limits; Sales cannot approve credit release; Finance can perform authorized review; both allowed and denied actions tested

Dependencies: NFR-001; named users; role design; RULE-005

Validation: Design walkthrough; planned SAP tests TC-007;TC-008;TC-019

18. FR-015

ID: FR-015

Type: Functional

Statement: The proposed process/solution shall trace critical changes and understand report freshness and errors.

Rationale: Audit and usable reporting addresses PROB-001 and supports BR-001.

Source: SIM-IT-01 (role-play)

Priority: Should

Priority reason: Supports operation; interim documented control possible

Acceptance: For promise/block/billing changes capture actor, time, before/after and reason; reports show as-of time; a user can locate order owner without a workaround

Dependencies: NFR-002;NFR-003;NFR-004;NFR-005; logging scope; RULE-012

Validation: Design walkthrough; planned SAP tests TC-020;TC-021

19. NFR-001

ID: NFR-001

Type: Non-functional

Statement: Access permissions shall meet the stated acceptance condition.

Rationale: Unauthorized financial actions are unacceptable, so tolerance is zero.

Source: Finance/IT role-play/control analysis

Priority: Must

Priority reason: Unauthorized financial actions are unacceptable, so tolerance is zero.

Acceptance: Every allow/deny action in the role matrix must match the proposed role restriction; zero unauthorized credit releases in the defined test set.

Dependencies: FR-014; RULE-005

Validation: Role matrix review and positive/negative system tests

20. NFR-002

ID: NFR-002

Type: Non-functional

Statement: Auditability shall meet the stated acceptance condition.

Rationale: All sampled critical events must be explainable; this is coverage, not a retention-law claim.

Source: Finance/IT role-play/control analysis

Priority: Must

Priority reason: All sampled critical events must be explainable; this is coverage, not a retention-law claim.

Acceptance: For every sampled promise, block release and invoice correction, retain actor, timestamp, before/after and reason; retention duration requires policy decision.

Dependencies: FR-015; record retention decision

Validation: Inspect audit records and compare controlled changes

21. NFR-003

ID: NFR-003

Type: Non-functional

Statement: Reporting timeliness shall meet the stated acceptance condition.

Rationale: A daily meeting is the business use case; real-time integration is not justified yet.

Source: Customer Service/IT role-play/control analysis

Priority: Should

Priority reason: A daily meeting is the business use case; real-time integration is not justified yet.

Acceptance: Daily management snapshot available before the proposed 09:00 Europe/London operations review with as-of timestamp; stale snapshot displays a warning.

Dependencies: FR-013; schedule; timezone conversion

Validation: Delay refresh deliberately and inspect timestamp/warning

22. NFR-004

ID: NFR-004

Type: Non-functional

Statement: Usability shall meet the stated acceptance condition.

Rationale: One person per role is formative coverage, not a statistically representative usability study.

Source: All roles role-play/control analysis

Priority: Should

Priority reason: One person per role is formative coverage, not a statistically representative usability study.

Acceptance: Each of five proposed role representatives can locate an order, its owner and next action using the guide; record completion, errors and time before agreeing a timing target.

Dependencies: FR-002; FR-015; participants unavailable

Validation: Moderated tasks; no participants tested in this portfolio

23. NFR-005

ID: NFR-005

Type: Non-functional

Statement: Data quality and reconciliation shall meet the stated acceptance condition.

Rationale: Exact arithmetic is appropriate for keys and ledger-like reconciliation; incomplete business data is separately flagged.

Source: Finance/IT role-play/control analysis

Priority: Must

Priority reason: Exact arithmetic is appropriate for keys and ledger-like reconciliation; incomplete business data is separately flagged.

Acceptance: Required keys are unique/non-null, child references valid, quantities and allocations within parent bounds, and integer-pence balances reconcile exactly.

Dependencies: FR-013; data dictionary

Validation: Executed Python/SQLite checks; SAP extraction validation still required

Appendix — Rules

Editable source: phase-04/business-rules.csv

1. RULE-001

ID: RULE-001

Rule: Before confirmation require active customer, ship-to, product, positive quantity, valid price, currency and requested date. Drafts may be incomplete.

Owner: Sales

Rationale: Avoid knowingly releasing incomplete orders; field list proposed, not a SAP default.

Status: Proposed simulation policy

2. RULE-002

ID: RULE-002

Rule: Every open exception needs one accountable role, next action and next review date.

Owner: Customer Service

Rationale: Daily review proposed to prevent unowned handoffs.

Status: Proposed simulation policy

3. RULE-003

ID: RULE-003

Rule: Keep requested receipt and initial confirmed receipt dates separately; later changes must not overwrite the original promise.

Owner: Sales

Rationale: Prevents improving on-time delivery by moving the denominator date.

Status: Proposed simulation policy

4. RULE-004

ID: RULE-004

Rule: Partial deliveries require customer agreement recorded on the order; default to review when unknown.

Owner: Customer Service

Rationale: A partial shipment can add cost or disrupt customer operations.

Status: Proposed simulation policy

5. RULE-005

ID: RULE-005

Rule: Credit release is a Finance responsibility; Sales and Warehouse may request review but not self-release. Manual order blocks retain their own reasons.

Owner: Finance

Rationale: Separation of duties; actual authorization model unvalidated.

Status: Proposed simulation policy

6. RULE-006

ID: RULE-006

Rule: Capture dispatch, goods-issue posting and receipt as separate events. Never treat goods issue as proof of customer receipt.

Owner: Warehouse

Rationale: Correct service measurement and posting reconciliation.

Status: Proposed simulation policy

7. RULE-007

ID: RULE-007

Rule: Cancel before dispatch only after checking dependent documents; after dispatch route to controlled return/reversal review.

Owner: Sales

Rationale: Avoid deleting history; full returns redesign is out of scope.

Status: Proposed simulation policy

8. RULE-008

ID: RULE-008

Rule: Finance reviews billable and unbilled exceptions each business day; escalate items still unbilled on the next review.

Owner: Finance

Rationale: Proposed daily operating cadence, not a proven 24-hour SAP service level.

Status: Proposed simulation policy

9. RULE-009

ID: RULE-009

Rule: Bill only eligible quantity once; preserve correction links and original posted documents.

Owner: Finance

Rationale: Invoice correctness and retry safety; exact system configuration unresolved.

Status: Proposed simulation policy

10. RULE-010

ID: RULE-010

Rule: Allocate only to supported customer/invoice matches; preserve partial and unallocated balances without automatic write-off.

Owner: Finance

Rationale: No invented tolerance or forced clearing.

Status: Proposed simulation policy

11. RULE-011

ID: RULE-011

Rule: Aggregate child records to their intended parent grain before combining measures.

Owner: Analyst

Rationale: Prevents delivery/payment join duplication.

Status: Proposed simulation policy

12. RULE-012

ID: RULE-012

Rule: Show report as-of time and preserve critical change history; do not present missing evidence as zero.

Owner: IT

Rationale: Allows users to assess reliability.

Status: Proposed simulation policy

Appendix — Traceability

Editable source: phase-04/traceability.csv

1. PROB-001

Problem: PROB-001

Business requirement: BR-001

Requirement: FR-001

Process change: PC-001

Process change description: Customer data validation

Story: US-001

Tests: TC-001;TC-002;TC-003

KPI: KPI-005

Rule: RULE-001

2. PROB-001

Problem: PROB-001

Business requirement: BR-001

Requirement: FR-002

Process change: PC-002

Process change description: Order status visibility

Story: US-002

Tests: TC-017;TC-021

KPI: KPI-006

Rule: RULE-002

3. PROB-002

Problem: PROB-002

Business requirement: BR-002

Requirement: FR-003

Process change: PC-003

Process change description: Availability-based promises

Story: US-003

Tests: TC-001;TC-004

KPI: KPI-002

Rule: RULE-003

4. PROB-002

Problem: PROB-002

Business requirement: BR-002

Requirement: FR-004

Process change: PC-004

Process change description: Partial delivery agreement

Story: US-004

Tests: TC-005;TC-006

KPI: KPI-003

Rule: RULE-004

5. PROB-002

Problem: PROB-002

Business requirement: BR-002

Requirement: FR-005

Process change: PC-005

Process change description: Credit and order blocks

Story: US-005

Tests: TC-007;TC-008

KPI: KPI-002

Rule: RULE-005

6. PROB-002

Problem: PROB-002

Business requirement: BR-002

Requirement: FR-006

Process change: PC-006

Process change description: Fulfilment and posting evidence

Story: US-006

Tests: TC-001;TC-005;TC-011

KPI: KPI-001

Rule: RULE-006

7. PROB-001

Problem: PROB-001

Business requirement: BR-001

Requirement: FR-007

Process change: PC-007

Process change description: Controlled cancellation

Story: US-007

Tests: TC-009;TC-010

KPI: KPI-003

Rule: RULE-007

8. PROB-003

Problem: PROB-003

Business requirement: BR-003

Requirement: FR-008

Process change: PC-008

Process change description: Billing readiness queue

Story: US-008

Tests: TC-001;TC-012;TC-022

KPI: KPI-004

Rule: RULE-008

9. PROB-003

Problem: PROB-003

Business requirement: BR-003

Requirement: FR-009

Process change: PC-009

Process change description: Invoice creation and duplicate control

Story: US-009

Tests: TC-001;TC-012;TC-013;TC-022

KPI: KPI-004

Rule: RULE-009

10. PROB-003

Problem: PROB-003

Business requirement: BR-003

Requirement: FR-010

Process change: PC-010

Process change description: Invoice correction

Story: US-010

Tests: TC-013

KPI: KPI-005

Rule: RULE-009

11. PROB-001

Problem: PROB-001

Business requirement: BR-001

Requirement: FR-011

Process change: PC-011

Process change description: Payment allocation

Story: US-011

Tests: TC-001;TC-014;TC-015;TC-016

KPI: Control evidence; no direct KPI

Rule: RULE-010

12. PROB-001

Problem: PROB-001

Business requirement: BR-001

Requirement: FR-012

Process change: PC-012

Process change description: Exception ownership

Story: US-012

Tests: TC-017

KPI: KPI-006

Rule: RULE-002

13. PROB-001

Problem: PROB-001

Business requirement: BR-001

Requirement: FR-013

Process change: PC-013

Process change description: KPI definitions and reconciliation

Story: US-013

Tests: TC-018

KPI: KPI-001;KPI-002;KPI-003;KPI-004;KPI-005;KPI-006

Rule: RULE-011

14. PROB-001

Problem: PROB-001

Business requirement: BR-001

Requirement: FR-014

Process change: PC-014

Process change description: Role-based access

Story: US-014

Tests: TC-007;TC-008;TC-019

KPI: Control evidence; no direct KPI

Rule: RULE-005

15. PROB-001

Problem: PROB-001

Business requirement: BR-001

Requirement: FR-015

Process change: PC-015

Process change description: Audit and usable reporting

Story: US-015

Tests: TC-020;TC-021

KPI: KPI-006

Rule: RULE-012

Appendix — Non-functional test links

Editable source: phase-04/nonfunctional-trace.csv

1. NFR-001

Requirement: NFR-001

Functional anchor: FR-014

Story: US-014

Tests: TC-007;TC-008;TC-019

Validation status: SAP not executed; NFR-005 synthetic subset executed

2. NFR-002

Requirement: NFR-002

Functional anchor: FR-015

Story: US-015

Tests: TC-020

Validation status: SAP not executed; NFR-005 synthetic subset executed

3. NFR-003

Requirement: NFR-003

Functional anchor: FR-015

Story: US-015

Tests: TC-020

Validation status: SAP not executed; NFR-005 synthetic subset executed

4. NFR-004

Requirement: NFR-004

Functional anchor: FR-015

Story: US-015

Tests: TC-021

Validation status: SAP not executed; NFR-005 synthetic subset executed

5. NFR-005

Requirement: NFR-005

Functional anchor: FR-013

Story: US-013

Tests: TC-018

Validation status: SAP not executed; NFR-005 synthetic subset executed

Appendix — Access matrix

Editable source: phase-04/access-matrix.csv

1. View operational status

Action: View operational status

Sales: Allow

Warehouse: Allow

Finance: Allow

Customer Service: Allow

IT: Deny

2. Confirm commercial promise

Action: Confirm commercial promise

Sales: Allow

Warehouse: Deny

Finance: Deny

Customer Service: Deny

IT: Deny

3. Request credit release

Action: Request credit release

Sales: Allow

Warehouse: Deny

Finance: Deny

Customer Service: Allow

IT: Deny

4. Approve credit release

Action: Approve credit release

Sales: Deny

Warehouse: Deny

Finance: Allow

Customer Service: Deny

IT: Deny

5. Post goods issue

Action: Post goods issue

Sales: Deny

Warehouse: Allow

Finance: Deny

Customer Service: Deny

IT: Deny

6. Create/correct invoice

Action: Create/correct invoice

Sales: Deny

Warehouse: Deny

Finance: Allow

Customer Service: Deny

IT: Deny

7. Allocate payment

Action: Allocate payment

Sales: Deny

Warehouse: Deny

Finance: Allow

Customer Service: Deny

IT: Deny

8. Change access configuration

Action: Change access configuration

Sales: Deny

Warehouse: Deny

Finance: Deny

Customer Service: Deny

IT: Allow

Appendix — Discovery findings

Editable source: phase-02/discovery-findings.csv

1. SIM-SALES-01

Source: SIM-SALES-01

Role: Sales

Role play statement: We sometimes learn about missing delivery details only when Warehouse asks.

Finding: Validation responsibility and required fields need definition.

Problem: PROB-001

Evidence status: Invented role-play; no interview conducted

2. SIM-SALES-02

Source: SIM-SALES-02

Role: Sales

Role play statement: Customers cancel, but we cannot always tell whether the order has already shipped.

Finding: Cancellation must check downstream state.

Problem: PROB-001

Evidence status: Invented role-play; no interview conducted

3. SIM-WH-01

Source: SIM-WH-01

Role: Warehouse

Role play statement: A shelf quantity is not a reliable promise if stock is already committed.

Finding: Availability and lead times must precede confirmation.

Problem: PROB-002

Evidence status: Invented role-play; no interview conducted

4. SIM-WH-02

Source: SIM-WH-02

Role: Warehouse

Role play statement: Dispatch paperwork and the recorded stock movement do not always happen together.

Finding: Keep physical and posting dates separate.

Problem: PROB-002

Evidence status: Invented role-play; no interview conducted

5. SIM-CS-01

Source: SIM-CS-01

Role: Customer Service

Role play statement: I ask several teams before I can explain a delay.

Finding: An order view and accountable exception owner are needed.

Problem: PROB-001

Evidence status: Invented role-play; no interview conducted

6. SIM-CS-02

Source: SIM-CS-02

Role: Customer Service

Role play statement: Some customers need part of an order urgently; others reject split shipments.

Finding: Record partial-delivery agreement; do not assume one rule for everyone.

Problem: PROB-002

Evidence status: Invented role-play; no interview conducted

7. SIM-FIN-01

Source: SIM-FIN-01

Role: Finance

Role play statement: Sales wants to release held orders, but Finance must own credit decisions.

Finding: Separate request and approval responsibilities.

Problem: PROB-002

Evidence status: Invented role-play; no interview conducted

8. SIM-FIN-02

Source: SIM-FIN-02

Role: Finance

Role play statement: We wait for dispatch emails before reviewing what should be billed.

Finding: Reconcile goods-issued deliveries with billing queue daily.

Problem: PROB-003

Evidence status: Invented role-play; no interview conducted

9. SIM-FIN-03

Source: SIM-FIN-03

Role: Finance

Role play statement: We need to explain price corrections without losing the first invoice.

Finding: Linked correction records are needed.

Problem: PROB-003

Evidence status: Invented role-play; no interview conducted

10. SIM-FIN-04

Source: SIM-FIN-04

Role: Finance

Role play statement: A receipt can be short or lack a usable invoice reference.

Finding: Retain partial and unallocated amounts.

Problem: PROB-001

Evidence status: Invented role-play; no interview conducted

11. SIM-IT-01

Source: SIM-IT-01

Role: IT

Role play statement: Joining several document lists may duplicate values; custom screens need a clear purpose.

Finding: Agree grains, freshness and standard-first fit.

Problem: PROB-001

Evidence status: Invented role-play; no interview conducted

Appendix — Responsibilities

Editable source: phase-02/responsibility-matrix.csv

1. Customer/order validation

Activity: Customer/order validation

Sales: A/R

Warehouse: I

Finance: C

Customer Service: C

IT: I

2. Promise and partial agreement

Activity: Promise and partial agreement

Sales: A/R

Warehouse: C

Finance: I

Customer Service: C

IT: I

3. Credit release

Activity: Credit release

Sales: C

Warehouse: I

Finance: A/R

Customer Service: I

IT: I

4. Dispatch and goods issue

Activity: Dispatch and goods issue

Sales: C

Warehouse: A/R

Finance: C

Customer Service: I

IT: I

5. Billing and corrections

Activity: Billing and corrections

Sales: I

Warehouse: C

Finance: A/R

Customer Service: C

IT: I

6. Payment allocation

Activity: Payment allocation

Sales: I

Warehouse: I

Finance: A/R

Customer Service: C

IT: I

7. Customer exception communication

Activity: Customer exception communication

Sales: C

Warehouse: C

Finance: C

Customer Service: A/R

IT: I

8. Role/report configuration

Activity: Role/report configuration

Sales: C

Warehouse: C

Finance: C

Customer Service: C

IT: A/R

Appendix — AS-IS step register

Editable source: phase-03/as-is-step-register.csv

1. AS-01

Step: AS-01

Owner: Sales

Input: Customer email/order

Output: Recorded request

Tool: Email and order spreadsheet

Rule control: Check customer reference manually

Pain point: Details rekeyed

Potential cause: No agreed required-field checklist

Evidence required: Original request vs entered fields

Consequence: Corrections and repeated contact

Evidence status: Fictional process assumption; not observed

2. AS-02

Step: AS-02

Owner: Sales / Finance

Input: Recorded request

Output: Validated or held order

Tool: Spreadsheet and messages

Rule control: Check customer, price and credit by request

Pain point: Hold reason not shared

Potential cause: Different teams keep separate lists

Evidence required: Rejected orders and hold logs

Consequence: Unclear release ownership

Evidence status: Fictional process assumption; not observed

3. AS-03

Step: AS-03

Owner: Sales / Warehouse

Input: Validated demand

Output: Informal promise or shortage query

Tool: Stock sheet and calls

Rule control: Ask about stock before picking

Pain point: Shelf count mistaken for available stock

Potential cause: Commitments and lead times not visible

Evidence required: Stock snapshot and concurrent commitments

Consequence: Unreliable date promise

Evidence status: Fictional process assumption; not observed

4. AS-04

Step: AS-04

Owner: Warehouse / Customer Service

Input: Pick request

Output: Picked goods or partial decision

Tool: Paper pick list and email

Rule control: Seek agreement for short shipment

Pain point: Partial permission hard to find

Potential cause: Agreement not attached to order

Evidence required: Customer correspondence and pick quantities

Consequence: Repeat handling or customer rejection

Evidence status: Fictional process assumption; not observed

5. AS-05

Step: AS-05

Owner: Warehouse

Input: Picked goods

Output: Dispatch note and stock movement entry

Tool: Paper note and stock workbook

Rule control: Record actual quantity and movement

Pain point: Physical event and entry occur at different times

Potential cause: Batch updates or missing owner

Evidence required: Dispatch log vs entry timestamps

Consequence: Status/stock mismatch

Evidence status: Fictional process assumption; not observed

6. AS-06

Step: AS-06

Owner: Finance

Input: Dispatch email

Output: Invoice or unresolved item

Tool: Standalone billing spreadsheet/tool

Rule control: Check price/quantity before sending

Pain point: Dispatch email awaited

Potential cause: No complete billable population reconciliation

Evidence required: Dispatched vs invoiced lists

Consequence: Late billing

Evidence status: Fictional process assumption; not observed

7. AS-07

Step: AS-07

Owner: Finance

Input: Bank receipt and remittance

Output: Allocation or unmatched balance

Tool: Bank file and receivables sheet

Rule control: Match customer/reference/amount

Pain point: Short payment handling inconsistent

Potential cause: No agreed partial/unmatched policy

Evidence required: Bank records and allocation history

Consequence: Misstated open balances

Evidence status: Fictional process assumption; not observed

8. AS-08

Step: AS-08

Owner: Customer Service / IT

Input: Departmental lists

Output: Order/KPI report

Tool: Excel consolidation

Rule control: Reconcile lists manually

Pain point: Different totals and stale status

Potential cause: One-to-many joins and inconsistent dates

Evidence required: Source extracts and formula review

Consequence: Poor operational decisions

Evidence status: Fictional process assumption; not observed

Appendix — SAP evidence

Editable source: phase-05/sap-evidence.csv

1. SAP-001

ID: SAP-001

Title: Understanding the Sell from Stock Process

URL: https://learning.sap.com/courses/implementing-sap-s-4hana-cloud-public-edition-sales-fundamental-business-processes/understanding-the-sell-from-stock-process_f487b5b8-429f-4f14-bacf-39d063789bad

Access date: 2026-10-11

Edition: SAP S/4HANA Cloud Public Edition

Release: 2602 reference; learning pages unpinned

Claim: BD9 connects sales order, outbound delivery, picking, goods issue and billing. Goods issue records stock movement and cost; billing supports revenue and receivables postings.

Locator: Sell from Stock Overview

Verification: Official page text reviewed

Limit: Learning page is not release-pinned; validate actual scope and posting behaviour in selected tenant.

System demonstration: None

2. SAP-002

ID: SAP-002

Title: Understanding Sales Functions

URL: https://learning.sap.com/courses/implementing-sap-s-4hana-cloud-public-edition-sales-fundamental-business-processes/understanding-sales-functions_c385afd5-c15f-431b-904a-a394fe7ed4e9

Access date: 2026-10-11

Edition: SAP S/4HANA Cloud Public Edition

Release: 2602 reference; learning pages unpinned

Claim: Availability and scheduling depend on master data and check settings; confirmed quantities and dates can differ from the requested date.

Locator: Delivery Scheduling; Availability Check

Verification: Official page text reviewed

Limit: No check scope or scheduling parameters configured or demonstrated.

System demonstration: None

3. SAP-003

ID: SAP-003

Title: Understanding Shipping Functions

URL: https://learning.sap.com/courses/implementing-sap-s-4hana-cloud-public-edition-sales-fundamental-business-processes/understanding-shipping-functions_b61d6a11-be4e-43ef-8a67-7d0c1738fb5c

Access date: 2026-10-11

Edition: SAP S/4HANA Cloud Public Edition

Release: 2602 reference; learning pages unpinned

Claim: Outbound deliveries support shipping. Complete/partial delivery settings can be maintained and carried into orders.

Locator: Processing Outbound Deliveries; Complete and Partial Delivery

Verification: Official page text reviewed

Limit: Customer agreement capture and exact restriction behaviour require validation.

System demonstration: None

4. SAP-004

ID: SAP-004

Title: Understanding Billing Functions

URL: https://learning.sap.com/courses/implementing-sap-s-4hana-cloud-public-edition-sales-fundamental-business-processes/understanding-billing-functions_a8a3c8d5-9ad5-4e81-ab36-4a550518137d

Access date: 2026-10-11

Edition: SAP S/4HANA Cloud Public Edition

Release: 2602 reference; learning pages unpinned

Claim: Billing due items can be processed manually or through scheduled jobs; accounts receivable follows billing in J59.

Locator: Processing Billing Documents; Billing Scheduling

Verification: Official page text reviewed

Limit: Job timing, errors, accounting release and output monitoring require tenant validation; no guaranteed automatic success.

System demonstration: None

5. SAP-005

ID: SAP-005

Title: Maintaining Business Partner Master Data

URL: https://learning.sap.com/courses/establishing-core-supply-chain-processes-in-sap-s-4hana-cloud-public-edition/maintaining-business-partner-master-data_c2bba12f-25e4-4ebb-8fc7-590eab92c621

Access date: 2026-10-11

Edition: SAP S/4HANA Cloud Public Edition

Release: 2602 reference; learning pages unpinned

Claim: The business partner model centrally maintains common data with role-specific customer information.

Locator: Business Partner Key Concepts

Verification: Official page text reviewed

Limit: Required fields, duplicate detection and approval design are project needs, not verified defaults.

System demonstration: None

6. SAP-006

ID: SAP-006

Title: Posting Manual Incoming Payments

URL: https://learning.sap.com/courses/customer-payments-in-receivables-management/posting-manual-incoming-payments_f94d8e0d-e36c-4efe-8779-bdf601c13106

Access date: 2026-10-11

Edition: SAP S/4HANA Cloud Public Edition

Release: 2602 reference; learning pages unpinned

Claim: Partial payments leave the original invoice open with a linked payment item; residual handling clears it and creates a remaining item. On-account posting is possible.

Locator: Manual Incoming Payments

Verification: Official page text reviewed

Limit: Mixed-edition course includes a Public Edition example. Learning content is not release-pinned. Allocation CSV is an analytical model, not SAP journal structure.

System demonstration: None

7. SAP-007

ID: SAP-007

Title: Public Edition 2602 Feature Scope Description

URL: https://help.sap.com/doc/7c9e0bbbd1664c2581b2038a1c7ae4b3/2602.500/en-US/FSD_CE2602.pdf

Access date: 2026-10-11

Edition: SAP S/4HANA Cloud Public Edition

Release: 2602, document v5.0 / 2026-06-11

Claim: Reference release 2602; document version 5.0, 11 June 2026. Role-based authorizations, order processing, credit checks and availability capabilities are described.

Locator: Printed pages 9–10, 103, 195–196, 264

Verification: Official release-specific PDF reviewed

Limit: Some features require separate subscriptions or landscape prerequisites. No entitlement or full custom control coverage inferred.

System demonstration: None

8. SAP-008

ID: SAP-008

Title: Sales Order object page

URL: https://help.sap.com/docs/SAP_S4HANA_CLOUD/a630d57fc5004c6383e7a81efee7a8bb/43336e5621eb9d24e10000000a44147b.html

Access date: 2026-10-11

Edition: SAP S/4HANA Cloud Public Edition

Release: 2602 reference; learning pages unpinned

Claim: Detailed status-view fit remains unverified.

Locator: Page returned no readable body through research tool

Verification: Unverified; do not use as proof

Limit: Search snippet alone is not the evidence standard used here.

System demonstration: None

9. SAP-009

ID: SAP-009

Title: Checking and Releasing SD Documents

URL: https://help.sap.com/docs/SAP_S4HANA_CLOUD/918bca53037f408f91a2295d04ac16bc/84452f7da9c4467b9228d049126a2907.html

Access date: 2026-10-11

Edition: SAP S/4HANA Cloud Public Edition

Release: 2602 reference; learning pages unpinned

Claim: Exact credit-release workflow remains unverified.

Locator: Page returned no readable body through research tool

Verification: Unverified; do not use as proof

Limit: Use SAP-007 only for broad credit capability; validate approvals and roles separately.

System demonstration: None

Appendix — Fit-gap

Editable source: phase-06/fit-gap.csv

1. FR-001

Requirement: FR-001

Required capability: prevent order confirmation until required customer and order fields are valid

Standard evidence: SAP-005

Assessment: Process/configuration change

Business impact: Customer data validation affects PROB-001

Proposed resolution: Define policy and demonstrate standard configuration first; do not assume default settings meet acceptance criteria.

Alternative: Documented manual control; standard view/export before custom development

Dependencies: Customer owner; approved field list

Validation remaining: Missing ship-to address or inactive customer prevents confirmation, identifies the field and preserves the draft; corrected data can be rechecked

Assessment type: Desk-based; no system demonstration

2. FR-002

Requirement: FR-002

Required capability: see quantities, dates, blocks, related documents and accountable owner in one order view

Standard evidence: SAP-008

Assessment: Unresolved

Business impact: Order status visibility affects PROB-001

Proposed resolution: Run a focused standard-process demonstration against acceptance criteria; retain controlled manual procedure pending evidence.

Alternative: Documented manual control; standard view/export before custom development

Dependencies: Document relationships; FR-003; NFR-003

Validation remaining: NS-ORD-1001 shows 100 ordered, availability unresolved, requested receipt 9 October and no invented confirmation; unknown facts are labelled

Assessment type: Desk-based; no system demonstration

3. FR-003

Requirement: FR-003

Required capability: check usable supply and commitments before promising quantity and date

Standard evidence: SAP-002

Assessment: Process/configuration change

Business impact: Availability-based promises affects PROB-002

Proposed resolution: Define policy and demonstrate standard configuration first; do not assume default settings meet acceptance criteria.

Alternative: Documented manual control; standard view/export before custom development

Dependencies: Stock accuracy; scheduling lead times

Validation remaining: 100 ordered with only 60 confirmed available cannot be promised as 100 on the requested date; confirmed quantity/date and check time are retained

Assessment type: Desk-based; no system demonstration

4. FR-004

Requirement: FR-004

Required capability: record customer agreement before splitting fulfilment

Standard evidence: SAP-003

Assessment: Process/configuration change

Business impact: Partial delivery agreement affects PROB-002

Proposed resolution: Define policy and demonstrate standard configuration first; do not assume default settings meet acceptance criteria.

Alternative: Documented manual control; standard view/export before custom development

Dependencies: FR-003; customer delivery policy

Validation remaining: With no partial-delivery permission, a 60/100 shipment is stopped for review; with recorded agreement, 60 can proceed and 40 remains open

Assessment type: Desk-based; no system demonstration

5. FR-005

Requirement: FR-005

Required capability: hold blocked orders until the appropriate authorized review

Standard evidence: SAP-007

Assessment: Process/configuration change

Business impact: Credit and order blocks affects PROB-002

Proposed resolution: Define policy and demonstrate standard configuration first; do not assume default settings meet acceptance criteria.

Alternative: Documented manual control; standard view/export before custom development

Dependencies: Credit limits; NFR-001; precise block configuration

Validation remaining: An unresolved credit or manual order block prevents dispatch release; Sales cannot release a credit block; Finance review records decision, reason and actor

Assessment type: Desk-based; no system demonstration

6. FR-006

Requirement: FR-006

Required capability: record actual pick, dispatch, goods-issue posting and customer receipt separately

Standard evidence: SAP-001;SAP-003

Assessment: Integration/extension candidate

Business impact: Fulfilment and posting evidence affects PROB-002

Proposed resolution: Try standard capture/export first; assess an integration only if the needed evidence cannot be captured reliably.

Alternative: Documented manual control; standard view/export before custom development

Dependencies: FR-004; delivery documents; receipt evidence

Validation remaining: Picked/shipped quantities do not exceed authorized remaining quantity; dispatch and posting dates remain distinct; receipt is never inferred from posting

Assessment type: Desk-based; no system demonstration

7. FR-007

Requirement: FR-007

Required capability: cancel unshipped demand without erasing transaction history

Standard evidence: No verified exact capability source

Assessment: Unresolved

Business impact: Controlled cancellation affects PROB-001

Proposed resolution: Run a focused standard-process demonstration against acceptance criteria; retain controlled manual procedure pending evidence.

Alternative: Documented manual control; standard view/export before custom development

Dependencies: Document status checks; returns/reversal dependency

Validation remaining: Pre-dispatch cancellation records reason and removes demand from release queue; post-dispatch request routes to Finance/Warehouse review without deleting posted records

Assessment type: Desk-based; no system demonstration

8. FR-008

Requirement: FR-008

Required capability: review eligible dispatched deliveries without waiting for emailed paperwork

Standard evidence: SAP-004

Assessment: Process/configuration change

Business impact: Billing readiness queue affects PROB-003

Proposed resolution: Define policy and demonstrate standard configuration first; do not assume default settings meet acceptance criteria.

Alternative: Documented manual control; standard view/export before custom development

Dependencies: FR-006; billing relevance; daily review calendar

Validation remaining: A goods-issued delivery enters the billing review population; blocked/incomplete entries show reason and owner; unbilled age remains visible

Assessment type: Desk-based; no system demonstration

9. FR-009

Requirement: FR-009

Required capability: bill eligible delivered quantities once and track posting/output status

Standard evidence: SAP-004

Assessment: Process/configuration change

Business impact: Invoice creation and duplicate control affects PROB-003

Proposed resolution: Define policy and demonstrate standard configuration first; do not assume default settings meet acceptance criteria.

Alternative: Documented manual control; standard view/export before custom development

Dependencies: FR-008; pricing/tax master data; document linkage

Validation remaining: For 60 brackets at £25, net billing is £1,500; retry must not create a second active charge for the same quantity; posting/output failures remain visible

Assessment type: Desk-based; no system demonstration

10. FR-010

Requirement: FR-010

Required capability: correct invoice errors through linked reviewed adjustments

Standard evidence: No verified exact capability source

Assessment: Unresolved

Business impact: Invoice correction affects PROB-003

Proposed resolution: Run a focused standard-process demonstration against acceptance criteria; retain controlled manual procedure pending evidence.

Alternative: Documented manual control; standard view/export before custom development

Dependencies: Correction method and tax treatment unresolved

Validation remaining: A wrong-price invoice is marked disputed and routed for review; correction preserves original reference and reason; no silent overwrite of a posted invoice

Assessment type: Desk-based; no system demonstration

11. FR-011

Requirement: FR-011

Required capability: match full and partial receipts while exposing unallocated balances

Standard evidence: SAP-006

Assessment: Supported in principle

Business impact: Payment allocation affects PROB-001

Proposed resolution: Use documented standard approach; demonstrate exact policy in a configured practice tenant before acceptance.

Alternative: Documented manual control; standard view/export before custom development

Dependencies: Customer reference; chosen partial vs residual policy

Validation remaining: Against £1,500 net-only illustrative invoice, £900 allocation leaves £600 analytical balance; unmatched £200 stays unallocated; allocation cannot exceed receipt or invoice balance

Assessment type: Desk-based; no system demonstration

12. FR-012

Requirement: FR-012

Required capability: assign each open exception to an owner and next action

Standard evidence: No verified exact capability source

Assessment: Unresolved

Business impact: Exception ownership affects PROB-001

Proposed resolution: Run a focused standard-process demonstration against acceptance criteria; retain controlled manual procedure pending evidence.

Alternative: Documented manual control; standard view/export before custom development

Dependencies: Role matrix; escalation policy

Validation remaining: Each open exception has order reference, category, accountable role and review date; overdue reviews appear in queue without exposing unauthorized finance detail

Assessment type: Desk-based; no system demonstration

13. FR-013

Requirement: FR-013

Required capability: calculate consistent measures without multiplying order values

Standard evidence: No verified exact capability source

Assessment: Integration/extension candidate

Business impact: KPI definitions and reconciliation affects PROB-001

Proposed resolution: Try standard capture/export first; assess an integration only if the needed evidence cannot be captured reliably.

Alternative: Documented manual control; standard view/export before custom development

Dependencies: Approved definitions; receipt capture; FR-006; FR-011

Validation remaining: Known fixture with two deliveries and two allocations retains one order value; cancelled/open/partial cases follow KPI contract; totals reconcile to source grains

Assessment type: Desk-based; no system demonstration

14. FR-014

Requirement: FR-014

Required capability: restrict sensitive actions to accountable roles

Standard evidence: SAP-007

Assessment: Process/configuration change

Business impact: Role-based access affects PROB-001

Proposed resolution: Define policy and demonstrate standard configuration first; do not assume default settings meet acceptance criteria.

Alternative: Documented manual control; standard view/export before custom development

Dependencies: NFR-001; named users; role design

Validation remaining: Warehouse cannot change credit limits; Sales cannot approve credit release; Finance can perform authorized review; both allowed and denied actions tested

Assessment type: Desk-based; no system demonstration

15. FR-015

Requirement: FR-015

Required capability: trace critical changes and understand report freshness and errors

Standard evidence: No verified exact capability source

Assessment: Unresolved

Business impact: Audit and usable reporting affects PROB-001

Proposed resolution: Run a focused standard-process demonstration against acceptance criteria; retain controlled manual procedure pending evidence.

Alternative: Documented manual control; standard view/export before custom development

Dependencies: NFR-002;NFR-003;NFR-004;NFR-005; logging scope

Validation remaining: For promise/block/billing changes capture actor, time, before/after and reason; reports show as-of time; a user can locate order owner without a workaround

Assessment type: Desk-based; no system demonstration

Appendix — User stories

Editable source: phase-07/backlog.csv

1. US-001

ID: US-001

Capability: Customer data validation

Requirement: FR-001

Role: Sales

Story: As Sales, I need to prevent order confirmation until required customer and order fields are valid, so that BR-001 is supported.

Business value: Reduce uncertainty and prevent lost exceptions.

Given: UT-02: ship-to blank; authorized role and relevant policy available.

When: Save draft; attempt confirmation; populate valid ship-to; revalidate

Then: Missing ship-to address or inactive customer prevents confirmation, identifies the field and preserves the draft; corrected data can be rechecked

Additional exception outcome: Draft retained; confirmation initially prevented with field-specific explanation; correction permits recheck

Priority: Must

Dependencies: Customer owner; approved field list

Exceptions: Missing or incorrect customer data

Tests: TC-001;TC-002;TC-003

Status: Proposed; not implemented in SAP

2. US-002

ID: US-002

Capability: Order status visibility

Requirement: FR-002

Role: Customer Service

Story: As Customer Service, I need to see quantities, dates, blocks, related documents and accountable owner in one order view, so that BR-001 is supported.

Business value: Reduce uncertainty and prevent lost exceptions.

Given: UT-17: NS-ORD-1001 unresolved stock; no receipt evidence; authorized role and relevant policy available.

When: Search order as Customer Service; inspect documents; assign owner/next review; simulate overdue review

Then: NS-ORD-1001 shows 100 ordered, availability unresolved, requested receipt 9 October and no invented confirmation; unknown facts are labelled

Additional exception outcome: 100 ordered; stock uncertainty visible; receipt unknown; overdue action surfaced; no invented confirmation

Priority: Must

Dependencies: Document relationships; FR-003; NFR-003

Exceptions: Partial delivery and stale status

Tests: TC-017;TC-021

Status: Proposed; not implemented in SAP

3. US-003

ID: US-003

Capability: Availability-based promises

Requirement: FR-003

Role: Sales

Story: As Sales, I need to check usable supply and commitments before promising quantity and date, so that BR-002 is supported.

Business value: Avoid unsupported promises and unauthorized shipments.

Given: UT-04: NS-ORD-1001 variant, 100 ordered, 60 available; authorized role and relevant policy available.

When: Run availability review; attempt full promise; record feasible quantity/date

Then: 100 ordered with only 60 confirmed available cannot be promised as 100 on the requested date; confirmed quantity/date and check time are retained

Additional exception outcome: Full requested promise not falsely confirmed; shortfall and proposed date visible

Priority: Must

Dependencies: Stock accuracy; scheduling lead times

Exceptions: Insufficient stock

Tests: TC-001;TC-004

Status: Proposed; not implemented in SAP

4. US-004

ID: US-004

Capability: Partial delivery agreement

Requirement: FR-004

Role: Customer Service

Story: As Customer Service, I need to record customer agreement before splitting fulfilment, so that BR-002 is supported.

Business value: Avoid unsupported promises and unauthorized shipments.

Given: UT-05: 100 ordered, permission recorded, 60 available; authorized role and relevant policy available.

When: Release 60; record dispatch and receipt; inspect remaining demand

Then: With no partial-delivery permission, a 60/100 shipment is stopped for review; with recorded agreement, 60 can proceed and 40 remains open

Additional exception outcome: 60 fulfilled and 40 open; agreement retained; order not marked fully fulfilled

Priority: Must

Dependencies: FR-003; customer delivery policy

Exceptions: Partial delivery

Tests: TC-005;TC-006

Status: Proposed; not implemented in SAP

5. US-005

ID: US-005

Capability: Credit and order blocks

Requirement: FR-005

Role: Finance

Story: As Finance, I need to hold blocked orders until the appropriate authorized review, so that BR-002 is supported.

Business value: Avoid unsupported promises and unauthorized shipments.

Given: UT-07: customer fails configured credit check; authorized role and relevant policy available.

When: Attempt dispatch; review as Finance; record justified release; recheck downstream eligibility

Then: An unresolved credit or manual order block prevents dispatch release; Sales cannot release a credit block; Finance review records decision, reason and actor

Additional exception outcome: Dispatch held until authorized decision; release history retained; other blocks still respected

Priority: Must

Dependencies: Credit limits; NFR-001; precise block configuration

Exceptions: Credit or order block

Tests: TC-007;TC-008

Status: Proposed; not implemented in SAP

6. US-006

ID: US-006

Capability: Fulfilment and posting evidence

Requirement: FR-006

Role: Warehouse

Story: As Warehouse, I need to record actual pick, dispatch, goods-issue posting and customer receipt separately, so that BR-002 is supported.

Business value: Avoid unsupported promises and unauthorized shipments.

Given: UT-05: 100 ordered, permission recorded, 60 available; authorized role and relevant policy available.

When: Release 60; record dispatch and receipt; inspect remaining demand

Then: Picked/shipped quantities do not exceed authorized remaining quantity; dispatch and posting dates remain distinct; receipt is never inferred from posting

Additional exception outcome: 60 fulfilled and 40 open; agreement retained; order not marked fully fulfilled

Priority: Must

Dependencies: FR-004; delivery documents; receipt evidence

Exceptions: Quantity mismatch; missing receipt

Tests: TC-001;TC-005;TC-011

Status: Proposed; not implemented in SAP

7. US-007

ID: US-007

Capability: Controlled cancellation

Requirement: FR-007

Role: Sales

Story: As Sales, I need to cancel unshipped demand without erasing transaction history, so that BR-001 is supported.

Business value: Reduce uncertainty and prevent lost exceptions.

Given: UT-09: unshipped order with draft delivery; authorized role and relevant policy available.

When: Request cancellation; inspect dependent delivery; cancel/reject through controlled route

Then: Pre-dispatch cancellation records reason and removes demand from release queue; post-dispatch request routes to Finance/Warehouse review without deleting posted records

Additional exception outcome: No remaining dispatch release or active demand; reason/history retained; no inappropriate stock posting

Priority: Must

Dependencies: Document status checks; returns/reversal dependency

Exceptions: Order cancellation

Tests: TC-009;TC-010

Status: Proposed; not implemented in SAP

8. US-008

ID: US-008

Capability: Billing readiness queue

Requirement: FR-008

Role: Finance

Story: As Finance, I need to review eligible dispatched deliveries without waiting for emailed paperwork, so that BR-003 is supported.

Business value: Remove avoidable handoff waiting without billing incorrectly.

Given: UT-12: 60 goods-issued brackets, £25 each; authorized role and relevant policy available.

When: Review due queue without dispatch email; bill once; retry same quantity

Then: A goods-issued delivery enters the billing review population; blocked/incomplete entries show reason and owner; unbilled age remains visible

Additional exception outcome: £1,500 net-only charge once; second active charge prevented; status is reviewable

Priority: Must

Dependencies: FR-006; billing relevance; daily review calendar

Exceptions: Late or incomplete handoff

Tests: TC-001;TC-012;TC-022

Status: Proposed; not implemented in SAP

9. US-009

ID: US-009

Capability: Invoice creation and duplicate control

Requirement: FR-009

Role: Finance

Story: As Finance, I need to bill eligible delivered quantities once and track posting/output status, so that BR-003 is supported.

Business value: Remove avoidable handoff waiting without billing incorrectly.

Given: UT-12: 60 goods-issued brackets, £25 each; authorized role and relevant policy available.

When: Review due queue without dispatch email; bill once; retry same quantity

Then: For 60 brackets at £25, net billing is £1,500; retry must not create a second active charge for the same quantity; posting/output failures remain visible

Additional exception outcome: £1,500 net-only charge once; second active charge prevented; status is reviewable

Priority: Must

Dependencies: FR-008; pricing/tax master data; document linkage

Exceptions: Duplicate billing; posting/output failure

Tests: TC-001;TC-012;TC-013;TC-022

Status: Proposed; not implemented in SAP

10. US-010

ID: US-010

Capability: Invoice correction

Requirement: FR-010

Role: Finance

Story: As Finance, I need to correct invoice errors through linked reviewed adjustments, so that BR-003 is supported.

Business value: Remove avoidable handoff waiting without billing incorrectly.

Given: UT-13: invoice wrongly priced £30 instead of £25 for 60 units; authorized role and relevant policy available.

When: Report dispute; seek Finance approval; execute validated correction route; review linked documents

Then: A wrong-price invoice is marked disputed and routed for review; correction preserves original reference and reason; no silent overwrite of a posted invoice

Additional exception outcome: Original retained; £300 net difference explained through linked correction; no silent overwrite

Priority: Must

Dependencies: Correction method and tax treatment unresolved

Exceptions: Incorrect invoice

Tests: TC-013

Status: Proposed; not implemented in SAP

11. US-011

ID: US-011

Capability: Payment allocation

Requirement: FR-011

Role: Finance

Story: As Finance, I need to match full and partial receipts while exposing unallocated balances, so that BR-001 is supported.

Business value: Reduce uncertainty and prevent lost exceptions.

Given: UT-14: £1,500 net-only invoice; supported receipt £900; authorized role and relevant policy available.

When: Select partial-payment policy; allocate/reference receipt; inspect remaining exposure and open items

Then: Against £1,500 net-only illustrative invoice, £900 allocation leaves £600 analytical balance; unmatched £200 stays unallocated; allocation cannot exceed receipt or invoice balance

Additional exception outcome: Analytical balance £600; do not claim original SAP invoice cleared under partial-payment treatment

Priority: Must

Dependencies: Customer reference; chosen partial vs residual policy

Exceptions: Partial or unmatched payment

Tests: TC-001;TC-014;TC-015;TC-016

Status: Proposed; not implemented in SAP

12. US-012

ID: US-012

Capability: Exception ownership

Requirement: FR-012

Role: Customer Service

Story: As Customer Service, I need to assign each open exception to an owner and next action, so that BR-001 is supported.

Business value: Reduce uncertainty and prevent lost exceptions.

Given: UT-17: NS-ORD-1001 unresolved stock; no receipt evidence; authorized role and relevant policy available.

When: Search order as Customer Service; inspect documents; assign owner/next review; simulate overdue review

Then: Each open exception has order reference, category, accountable role and review date; overdue reviews appear in queue without exposing unauthorized finance detail

Additional exception outcome: 100 ordered; stock uncertainty visible; receipt unknown; overdue action surfaced; no invented confirmation

Priority: Should

Dependencies: Role matrix; escalation policy

Exceptions: Unowned block; overdue follow-up

Tests: TC-017

Status: Proposed; not implemented in SAP

13. US-013

ID: US-013

Capability: KPI definitions and reconciliation

Requirement: FR-013

Role: Analyst

Story: As Analyst, I need to calculate consistent measures without multiplying order values, so that BR-001 is supported.

Business value: Reduce uncertainty and prevent lost exceptions.

Given: UT-18: FIX-01–FIX-04 in quality/fixtures.json; authorized role and relevant policy available.

When: Calculate six measures; add second delivery and two allocations; compare with hand calculation

Then: Known fixture with two deliveries and two allocations retains one order value; cancelled/open/partial cases follow KPI contract; totals reconcile to source grains

Additional exception outcome: Matches fixture oracle; child fan-out does not inflate order value; cancelled/open/partial treatment follows KPI contract

Priority: Must

Dependencies: Approved definitions; receipt capture; FR-006; FR-011

Exceptions: Fan-out; missing dates

Tests: TC-018

Status: Proposed; not implemented in SAP

14. US-014

ID: US-014

Capability: Role-based access

Requirement: FR-014

Role: IT

Story: As IT, I need to restrict sensitive actions to accountable roles, so that BR-001 is supported.

Business value: Reduce uncertainty and prevent lost exceptions.

Given: UT-07: customer fails configured credit check; authorized role and relevant policy available.

When: Attempt dispatch; review as Finance; record justified release; recheck downstream eligibility

Then: Warehouse cannot change credit limits; Sales cannot approve credit release; Finance can perform authorized review; both allowed and denied actions tested

Additional exception outcome: Dispatch held until authorized decision; release history retained; other blocks still respected

Priority: Must

Dependencies: NFR-001; named users; role design

Exceptions: Unauthorized release or master-data edit

Tests: TC-007;TC-008;TC-019

Status: Proposed; not implemented in SAP

15. US-015

ID: US-015

Capability: Audit and usable reporting

Requirement: FR-015

Role: Customer Service

Story: As Customer Service, I need to trace critical changes and understand report freshness and errors, so that BR-001 is supported.

Business value: Reduce uncertainty and prevent lost exceptions.

Given: UT-20: change initial promise display, release block, invoice correction; stale snapshot; authorized role and relevant policy available.

When: Make authorized changes; inspect history; delay refresh beyond proposed daily review

Then: For promise/block/billing changes capture actor, time, before/after and reason; reports show as-of time; a user can locate order owner without a workaround

Additional exception outcome: Original promise preserved; actor/time/reason/before-after retained; stale as-of warning displayed

Priority: Should

Dependencies: NFR-002;NFR-003;NFR-004;NFR-005; logging scope

Exceptions: Missing audit trail; stale or inaccessible view

Tests: TC-020;TC-021

Status: Proposed; not implemented in SAP

Appendix — UAT case designs

Editable source: phase-09/uat-cases.csv

1. TC-001

ID: TC-001

Requirements: FR-001;FR-003;FR-006;FR-008;FR-009;FR-011

Scenario: Normal stock sale

Purpose: Validate FR-001, FR-003, FR-006, FR-008, FR-009, FR-011 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-01: 100 brackets, £25, valid customer, 100 available, no block

Steps: 1. Create and validate order; 2. confirm date; 3. create delivery; 4. pick/dispatch 100; 5. post goods issue; 6. bill; 7. allocate full supported payment

Expected results: Linked documents and quantities reconcile; separate event dates retained; no open analytical invoice balance

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

2. TC-002

ID: TC-002

Requirements: FR-001

Scenario: Missing customer information

Purpose: Validate FR-001 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-02: ship-to blank

Steps: 1. Save draft; 2. attempt confirmation; 3. populate valid ship-to; 4. revalidate

Expected results: Draft retained; confirmation initially prevented with field-specific explanation; correction permits recheck

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

3. TC-003

ID: TC-003

Requirements: FR-001

Scenario: Incorrect customer information

Purpose: Validate FR-001 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-03: inactive customer with populated address

Steps: 1. Attempt confirmation; 2. request customer-master correction; 3. revalidate after controlled correction

Expected results: Populated but invalid record is rejected; no bypass merely because fields are non-empty

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

4. TC-004

ID: TC-004

Requirements: FR-003

Scenario: Insufficient stock

Purpose: Validate FR-003 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-04: NS-ORD-1001 variant, 100 ordered, 60 available

Steps: 1. Run availability review; 2. attempt full promise; 3. record feasible quantity/date

Expected results: Full requested promise not falsely confirmed; shortfall and proposed date visible

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

5. TC-005

ID: TC-005

Requirements: FR-004;FR-006

Scenario: Partial delivery allowed

Purpose: Validate FR-004, FR-006 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-05: 100 ordered, permission recorded, 60 available

Steps: 1. Release 60; 2. record dispatch and receipt; 3. inspect remaining demand

Expected results: 60 fulfilled and 40 open; agreement retained; order not marked fully fulfilled

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

6. TC-006

ID: TC-006

Requirements: FR-004

Scenario: Partial delivery not allowed

Purpose: Validate FR-004 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-06: 100 ordered, permission denied, 60 available

Steps: 1. Attempt release of 60; 2. inspect outcome

Expected results: Split release prevented or routed for explicit decision; no dispatch authorization without agreement

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

7. TC-007

ID: TC-007

Requirements: FR-005;FR-014;NFR-001

Scenario: Credit block and authorized release

Purpose: Validate FR-005, FR-014 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-07: customer fails configured credit check

Steps: 1. Attempt dispatch; 2. review as Finance; 3. record justified release; 4. recheck downstream eligibility

Expected results: Dispatch held until authorized decision; release history retained; other blocks still respected

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

8. TC-008

ID: TC-008

Requirements: FR-005;FR-014;NFR-001

Scenario: Unauthorized and manual-block bypass

Purpose: Validate FR-005, FR-014 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-08: Sales user, credit block and separate manual order block

Steps: 1. Try credit release as Sales; 2. have Finance release credit; 3. attempt dispatch while manual block remains

Expected results: Sales release denied; manual block still prevents dispatch after credit release

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

9. TC-009

ID: TC-009

Requirements: FR-007

Scenario: Cancellation before dispatch

Purpose: Validate FR-007 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-09: unshipped order with draft delivery

Steps: 1. Request cancellation; 2. inspect dependent delivery; 3. cancel/reject through controlled route

Expected results: No remaining dispatch release or active demand; reason/history retained; no inappropriate stock posting

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

10. TC-010

ID: TC-010

Requirements: FR-007

Scenario: Cancellation after dispatch

Purpose: Validate FR-007 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-10: goods-issued order

Steps: 1. Request cancellation; 2. attempt record deletion; 3. route to Finance/Warehouse review

Expected results: No silent deletion or stock restoration; controlled return/reversal dependency raised

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

11. TC-011

ID: TC-011

Requirements: FR-006

Scenario: Quantity mismatch and posting timing

Purpose: Validate FR-006 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-11: 60 authorized, 55 physically picked, dispatch Monday, posting Tuesday

Steps: 1. Attempt posting 60; 2. correct quantity; 3. record distinct dates; 4. inspect outstanding demand

Expected results: Mismatch resolved before posting; remaining quantity accurate; dispatch not overwritten by posting date

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

12. TC-012

ID: TC-012

Requirements: FR-008;FR-009

Scenario: Billing handoff and duplicate retry

Purpose: Validate FR-008, FR-009 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-12: 60 goods-issued brackets, £25 each

Steps: 1. Review due queue without dispatch email; 2. bill once; 3. retry same quantity

Expected results: £1,500 net-only charge once; second active charge prevented; status is reviewable

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

13. TC-013

ID: TC-013

Requirements: FR-009;FR-010

Scenario: Incorrect invoice correction

Purpose: Validate FR-009, FR-010 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-13: invoice wrongly priced £30 instead of £25 for 60 units

Steps: 1. Report dispute; 2. seek Finance approval; 3. execute validated correction route; 4. review linked documents

Expected results: Original retained; £300 net difference explained through linked correction; no silent overwrite

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

14. TC-014

ID: TC-014

Requirements: FR-011

Scenario: Partial payment

Purpose: Validate FR-011 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-14: £1,500 net-only invoice; supported receipt £900

Steps: 1. Select partial-payment policy; 2. allocate/reference receipt; 3. inspect remaining exposure and open items

Expected results: Analytical balance £600; do not claim original SAP invoice cleared under partial-payment treatment

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

15. TC-015

ID: TC-015

Requirements: FR-011

Scenario: Unmatched payment

Purpose: Validate FR-011 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-15: £200 receipt with unknown invoice

Steps: 1. Attempt match; 2. leave on account/unallocated pending investigation; 3. assign owner

Expected results: No forced invoice allocation; £200 remains visible and traceable for Finance review

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

16. TC-016

ID: TC-016

Requirements: FR-011

Scenario: Over-allocation prevention

Purpose: Validate FR-011 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-16: £900 receipt; £600 invoice balance

Steps: 1. Attempt allocating £1,000 then £700; 2. correct to £600

Expected results: Both invalid attempts rejected; £300 receipt remains unallocated after valid allocation

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

17. TC-017

ID: TC-017

Requirements: FR-002;FR-012

Scenario: Order visibility and exception ownership

Purpose: Validate FR-002, FR-012 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-17: NS-ORD-1001 unresolved stock; no receipt evidence

Steps: 1. Search order as Customer Service; 2. inspect documents; 3. assign owner/next review; 4. simulate overdue review

Expected results: 100 ordered; stock uncertainty visible; receipt unknown; overdue action surfaced; no invented confirmation

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

18. TC-018

ID: TC-018

Requirements: FR-013;NFR-005

Scenario: KPI grain and denominator boundary

Purpose: Validate FR-013 including negative/exception behaviour

Preconditions: Agreed KPI definitions and fixture data available.

Test data: UT-18: FIX-01–FIX-04 in quality/fixtures.json

Steps: 1. Calculate six measures; 2. add second delivery and two allocations; 3. compare with hand calculation

Expected results: Matches fixture oracle; child fan-out does not inflate order value; cancelled/open/partial treatment follows KPI contract

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

19. TC-019

ID: TC-019

Requirements: FR-014;NFR-001

Scenario: Role access matrix

Purpose: Validate FR-014 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-19: Sales, Warehouse, Finance, Customer Service, IT users

Steps: 1. Try permitted and forbidden actions per phase-04/access-matrix.csv

Expected results: All matrix decisions respected; Warehouse cannot change credit terms; access denial is clear

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

20. TC-020

ID: TC-020

Requirements: FR-015;NFR-002;NFR-003

Scenario: Audit trail and freshness

Purpose: Validate FR-015 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-20: change initial promise display, release block, invoice correction; stale snapshot

Steps: 1. Make authorized changes; 2. inspect history; 3. delay refresh beyond proposed daily review

Expected results: Original promise preserved; actor/time/reason/before-after retained; stale as-of warning displayed

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

21. TC-021

ID: TC-021

Requirements: FR-015;FR-002;NFR-004

Scenario: Usability by role

Purpose: Validate FR-015, FR-002 including negative/exception behaviour

Preconditions: Five consenting role representatives and a usable practice system available; no participants recruited.

Test data: UT-21: five role representatives; order-search task

Steps: 1. Provide guide; 2. ask each to find order owner and next action; 3. observe keyboard use; 4. record times/errors

Expected results: Every participant completes task or usability issue is recorded; no timing target invented after results

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

22. TC-022

ID: TC-022

Requirements: FR-009;FR-008

Scenario: Billing posting/output failure

Purpose: Validate FR-009, FR-008 including negative/exception behaviour

Preconditions: Configured non-production Public Edition reference process; valid synthetic customer/product and named role; relevant rules/authorizations and resettable test state available.

Test data: UT-22: eligible invoice with simulated accounting release or output error

Steps: 1. Attempt billing; 2. trigger controlled failure; 3. review exception; 4. correct and retry

Expected results: Creation, accounting and customer-output statuses remain distinct; failure has Finance owner; retry avoids duplicate charge

Actual results: Not run in SAP

Status: Not executed

Evidence reference: None — design only

Defect reference: None

Execution type: Planned SAP UAT / business acceptance

Appendix — KPI definitions

Editable source: phase-08/kpi-contract.csv

1. KPI-001

ID: KPI-001

Name: Order cycle time

Formula: Mean(full_customer_receipt_date − order_received_date)

Denominator: Fully received noncancelled orders in intake cohort

Source fields: received_date; customer_receipt_date; shipped_qty; ordered_qty

Owner: Warehouse

Purpose: End-to-end physical fulfilment speed

Exclusions and limitations: Cancelled and incomplete orders excluded; final partial receipt completes order. Survivorship bias: disclose 51 open noncancelled orders.

Date basis: Intake cohort 2026-07-01 through 2026-10-11; snapshot 2026-10-11; calendar days; dates interpreted as UK business-local dates

Business target: Unresolved; synthetic baseline is not a target

2. KPI-002

ID: KPI-002

Name: On-time delivery %

Formula: 100 × orders fully received by initial confirmed receipt date / orders with initial promise due by as-of

Denominator: Noncancelled orders with nonblank initial_confirmed_receipt_date ≤ as-of

Source fields: initial_confirmed_receipt_date; customer_receipt_date; shipped_qty; ordered_qty

Owner: Sales

Purpose: Reliability of original promise

Exclusions and limitations: Open overdue and partially fulfilled overdue orders fail; future promises excluded; unknown promise excluded and counted separately. Do not substitute requested date.

Date basis: Intake cohort 2026-07-01 through 2026-10-11; snapshot 2026-10-11; calendar days; dates interpreted as UK business-local dates

Business target: Unresolved; synthetic baseline is not a target

3. KPI-003

ID: KPI-003

Name: Order fulfilment rate %

Formula: 100 × received units / ordered units

Denominator: All noncancelled orders in intake cohort, including open and future-due

Source fields: ordered_qty; shipped_qty; customer_receipt_date

Owner: Warehouse

Purpose: Quantity fulfilment as of snapshot

Exclusions and limitations: Cancelled excluded; partial contributes received units only; dispatched but not received does not count. Mixed products all use EA but units have unequal value.

Date basis: Intake cohort 2026-07-01 through 2026-10-11; snapshot 2026-10-11; calendar days; dates interpreted as UK business-local dates

Business target: Unresolved; synthetic baseline is not a target

4. KPI-004

ID: KPI-004

Name: Billing cycle time

Formula: Mean(invoice_date − goods_issue_posting_date)

Denominator: Invoiced deliveries with goods issue by as-of

Source fields: invoice_date; goods_issue_posting_date; delivery_id

Owner: Finance

Purpose: Posting-to-billing delay

Exclusions and limitations: Unbilled deliveries excluded from mean but disclosed separately; partial deliveries measured individually; net cancellations/reversals absent from data. Not dispatch-to-billing time.

Date basis: Intake cohort 2026-07-01 through 2026-10-11; snapshot 2026-10-11; calendar days; dates interpreted as UK business-local dates

Business target: Unresolved; synthetic baseline is not a target

5. KPI-005

ID: KPI-005

Name: Order error rate %

Formula: 100 × distinct orders with validation_error event / received orders

Denominator: All 300 orders in intake cohort including later cancellations

Source fields: events.event_type; events.order_id; orders.received_date

Owner: Sales

Purpose: First-pass validation quality

Exclusions and limitations: Corrected errors still count once; open/partial/cancelled included. Does not measure undetected or invoice errors.

Date basis: Intake cohort 2026-07-01 through 2026-10-11; snapshot 2026-10-11; calendar days; dates interpreted as UK business-local dates

Business target: Unresolved; synthetic baseline is not a target

6. KPI-006

ID: KPI-006

Name: Manual-touch rate %

Formula: 100 × distinct orders with manual_touch event / received orders

Denominator: All 300 orders in intake cohort including later cancellations

Source fields: events.event_type; events.order_id; orders.received_date

Owner: Customer Service

Purpose: Avoidable intervention prevalence

Exclusions and limitations: Includes chase, rekey and handoff enquiry, excludes routine physical picks; multiple touches count once. Logging completeness and process stage affect rate.

Date basis: Intake cohort 2026-07-01 through 2026-10-11; snapshot 2026-10-11; calendar days; dates interpreted as UK business-local dates

Business target: Unresolved; synthetic baseline is not a target

Appendix — Data model

Editable source: phase-08/data-model.csv

1. customers

Table: customers

Grain: One fictional customer

Primary key: customer_id

Foreign keys: None

2. products

Table: products

Grain: One stocked product

Primary key: product_id

Foreign keys: None

3. orders

Table: orders

Grain: One single-line order

Primary key: order_id

Foreign keys: customer_id → customers; product_id → products

4. deliveries

Table: deliveries

Grain: One shipment for an order

Primary key: delivery_id

Foreign keys: order_id → orders

5. invoices

Table: invoices

Grain: One active net-only invoice per delivery

Primary key: invoice_id

Foreign keys: delivery_id → deliveries

6. receipts

Table: receipts

Grain: One incoming net-only receipt

Primary key: receipt_id

Foreign keys: customer_id → customers

7. allocations

Table: allocations

Grain: One receipt-to-invoice allocation

Primary key: allocation_id

Foreign keys: receipt_id → receipts; invoice_id → invoices

8. events

Table: events

Grain: One intake/quality/manual/cancellation event

Primary key: event_id

Foreign keys: order_id → orders

Appendix — Data dictionary

Editable source: phase-08/data-dictionary.csv

1. customers

Table: customers

Field: customer_id

Type: text

Nullable: No

Definition: Stable synthetic identifier

Example: CUST-001

Rule: Unique non-null primary key

2. customers

Table: customers

Field: customer_name

Type: text

Nullable: No

Definition: Fictional descriptive name

Example: Alder Engineering Ltd

Rule: See generation assumptions and KPI contract

3. customers

Table: customers

Field: country

Type: text

Nullable: No

Definition: GB only

Example: GB

Rule: See generation assumptions and KPI contract

4. customers

Table: customers

Field: currency

Type: text

Nullable: No

Definition: GBP only

Example: GBP

Rule: See generation assumptions and KPI contract

5. customers

Table: customers

Field: payment_terms_days

Type: integer

Nullable: No

Definition: 30 calendar days assumed

Example: 30

Rule: See generation assumptions and KPI contract

6. products

Table: products

Field: product_id

Type: text

Nullable: No

Definition: Stable synthetic identifier

Example: PROD-001

Rule: Unique non-null primary key

7. products

Table: products

Field: product_name

Type: text

Nullable: No

Definition: Fictional descriptive name

Example: Standard mounting bracket

Rule: See generation assumptions and KPI contract

8. products

Table: products

Field: unit_price_pence

Type: integer

Nullable: No

Definition: Net GBP unit price in integer pence

Example: 2500

Rule: See generation assumptions and KPI contract

9. products

Table: products

Field: uom

Type: text

Nullable: No

Definition: EA, each

Example: EA

Rule: See generation assumptions and KPI contract

10. orders

Table: orders

Field: order_id

Type: text

Nullable: No

Definition: Stable synthetic identifier

Example: NS-ORD-1001

Rule: Unique non-null primary key

11. orders

Table: orders

Field: customer_id

Type: text

Nullable: No

Definition: Stable synthetic identifier

Example: CUST-001

Rule: Referential integrity

12. orders

Table: orders

Field: product_id

Type: text

Nullable: No

Definition: Stable synthetic identifier

Example: PROD-001

Rule: Referential integrity

13. orders

Table: orders

Field: warehouse_id

Type: text

Nullable: No

Definition: WH-01 only

Example: WH-01

Rule: See generation assumptions and KPI contract

14. orders

Table: orders

Field: received_date

Type: ISO date

Nullable: No

Definition: Order/customer cash receipt date according to table; actual, not future

Example: 2026-10-05

Rule: See generation assumptions and KPI contract

15. orders

Table: orders

Field: requested_receipt_date

Type: ISO date

Nullable: No

Definition: Customer request; not a confirmed promise

Example: 2026-10-09

Rule: See generation assumptions and KPI contract

16. orders

Table: orders

Field: initial_confirmed_receipt_date

Type: ISO date

Nullable: Yes

Definition: Original supported receipt promise; blank means unknown

Rule: See generation assumptions and KPI contract

17. orders

Table: orders

Field: ordered_qty

Type: integer

Nullable: No

Definition: Positive ordered units in EA

Example: 100

Rule: See generation assumptions and KPI contract

18. orders

Table: orders

Field: unit_price_pence

Type: integer

Nullable: No

Definition: Net GBP unit price in integer pence

Example: 2500

Rule: See generation assumptions and KPI contract

19. orders

Table: orders

Field: net_value_pence

Type: integer

Nullable: No

Definition: Ordered quantity × net unit price

Example: 250000

Rule: See generation assumptions and KPI contract

20. orders

Table: orders

Field: cancelled

Type: integer

Nullable: No

Definition: 1 if cancelled before dispatch, else 0

Example: 0

Rule: See generation assumptions and KPI contract

21. orders

Table: orders

Field: cancelled_date

Type: ISO date

Nullable: Yes

Definition: Cancellation date; blank if not cancelled

Rule: See generation assumptions and KPI contract

22. orders

Table: orders

Field: partial_allowed

Type: integer

Nullable: No

Definition: 1 if partial permission exists in this synthetic baseline; else 0

Example: 0

Rule: See generation assumptions and KPI contract

23. orders

Table: orders

Field: exception

Type: text

Nullable: No

Definition: Synthetic explanatory label, not a measured root cause

Example: Stock availability unresolved

Rule: See generation assumptions and KPI contract

24. deliveries

Table: deliveries

Field: delivery_id

Type: text

Nullable: No

Definition: Stable synthetic identifier

Example: DEL-0001

Rule: Unique non-null primary key

25. deliveries

Table: deliveries

Field: order_id

Type: text

Nullable: No

Definition: Stable synthetic identifier

Example: NS-ORD-1002

Rule: Referential integrity

26. deliveries

Table: deliveries

Field: shipped_qty

Type: integer

Nullable: No

Definition: Dispatched quantity in EA

Example: 60

Rule: See generation assumptions and KPI contract

27. deliveries

Table: deliveries

Field: dispatch_date

Type: ISO date

Nullable: No

Definition: Physical departure date

Example: 2026-09-02

Rule: See generation assumptions and KPI contract

28. deliveries

Table: deliveries

Field: goods_issue_posting_date

Type: ISO date

Nullable: Yes

Definition: System posting date; blank if not posted by snapshot

Example: 2026-09-02

Rule: See generation assumptions and KPI contract

29. deliveries

Table: deliveries

Field: customer_receipt_date

Type: ISO date

Nullable: Yes

Definition: Physical receipt evidence date; blank means unknown/not yet observed

Example: 2026-09-04

Rule: See generation assumptions and KPI contract

30. invoices

Table: invoices

Field: invoice_id

Type: text

Nullable: No

Definition: Stable synthetic identifier

Example: INV-0001

Rule: Unique non-null primary key

31. invoices

Table: invoices

Field: delivery_id

Type: text

Nullable: No

Definition: Stable synthetic identifier

Example: DEL-0001

Rule: Referential integrity

32. invoices

Table: invoices

Field: invoice_date

Type: ISO date

Nullable: No

Definition: Invoice creation date in simplified net-only model

Example: 2026-09-05

Rule: See generation assumptions and KPI contract

33. invoices

Table: invoices

Field: due_date

Type: ISO date

Nullable: No

Definition: Invoice date + 30 calendar days; may lie after snapshot

Example: 2026-10-05

Rule: See generation assumptions and KPI contract

34. invoices

Table: invoices

Field: net_amount_pence

Type: integer

Nullable: No

Definition: Net-only invoice amount; VAT and accounting corrections omitted

Example: 240000

Rule: See generation assumptions and KPI contract

35. receipts

Table: receipts

Field: receipt_id

Type: text

Nullable: No

Definition: Stable synthetic identifier

Example: PAY-0001

Rule: Unique non-null primary key

36. receipts

Table: receipts

Field: customer_id

Type: text

Nullable: No

Definition: Stable synthetic identifier

Example: CUST-002

Rule: Referential integrity

37. receipts

Table: receipts

Field: received_date

Type: ISO date

Nullable: No

Definition: Order/customer cash receipt date according to table; actual, not future

Example: 2026-09-16

Rule: See generation assumptions and KPI contract

38. receipts

Table: receipts

Field: amount_pence

Type: integer

Nullable: No

Definition: Net-only cash simulation amount, not actual VAT-inclusive customer cash

Example: 240000

Rule: See generation assumptions and KPI contract

39. receipts

Table: receipts

Field: reference

Type: text

Nullable: No

Definition: Synthetic remittance reference; unmatched text retained

Example: INV-0001

Rule: See generation assumptions and KPI contract

40. allocations

Table: allocations

Field: allocation_id

Type: text

Nullable: No

Definition: Stable synthetic identifier

Example: ALLOC-0001

Rule: Unique non-null primary key

41. allocations

Table: allocations

Field: receipt_id

Type: text

Nullable: No

Definition: Stable synthetic identifier

Example: PAY-0001

Rule: Referential integrity

42. allocations

Table: allocations

Field: invoice_id

Type: text

Nullable: No

Definition: Stable synthetic identifier

Example: INV-0001

Rule: Referential integrity

43. allocations

Table: allocations

Field: allocation_date

Type: ISO date

Nullable: No

Definition: Date receipt assigned analytically to invoice

Example: 2026-09-16

Rule: See generation assumptions and KPI contract

44. allocations

Table: allocations

Field: allocated_pence

Type: integer

Nullable: No

Definition: Positive net-only amount assigned; not SAP clearing journal

Example: 240000

Rule: See generation assumptions and KPI contract

45. events

Table: events

Field: event_id

Type: text

Nullable: No

Definition: Stable synthetic identifier

Example: EVT-0001

Rule: Unique non-null primary key

46. events

Table: events

Field: order_id

Type: text

Nullable: No

Definition: Stable synthetic identifier

Example: NS-ORD-1001

Rule: Referential integrity

47. events

Table: events

Field: event_type

Type: text

Nullable: No

Definition: order_received, validation_error, manual_touch or cancelled

Example: order_received

Rule: See generation assumptions and KPI contract

48. events

Table: events

Field: event_date

Type: ISO date

Nullable: No

Definition: Observed synthetic event date

Example: 2026-10-05

Rule: See generation assumptions and KPI contract

49. events

Table: events

Field: role

Type: text

Nullable: No

Definition: Fictional role performing event

Example: Sales

Rule: See generation assumptions and KPI contract

50. events

Table: events

Field: detail

Type: text

Nullable: No

Definition: Synthetic event explanation

Example: Synthetic intake

Rule: See generation assumptions and KPI contract

Appendix — Risks

Editable source: phase-10/risks.csv

1. RISK-001

ID: RISK-001

Risk: Unverified stock accuracy leads to unreliable promises

Owner: Warehouse

Likelihood: 4

Impact: 4

Score: 16

Mitigation: Reconcile stock; distinguish on-hand from confirmable quantity

Trigger: Repeated pick shortage

Contingency: Stock review before confirming

Status: Open; illustrative assessment

2. RISK-002

ID: RISK-002

Risk: Poor customer data blocks or misdirects orders

Owner: Sales

Likelihood: 4

Impact: 3

Score: 12

Mitigation: Named master-data owner; field validation and correction queue

Trigger: Repeated missing ship-to

Contingency: Hold incomplete orders

Status: Open; illustrative assessment

3. RISK-003

ID: RISK-003

Risk: Manual handoff survives alongside new queue

Owner: Finance

Likelihood: 3

Impact: 4

Score: 12

Mitigation: Own daily reconciliation of goods-issued vs billed deliveries

Trigger: Unbilled age rises

Contingency: Manual documented daily reconciliation

Status: Open; illustrative assessment

4. RISK-004

ID: RISK-004

Risk: Users bypass credit segregation

Owner: Finance

Likelihood: 2

Impact: 5

Score: 10

Mitigation: Negative authorization tests and review of privileged access

Trigger: Unauthorized release attempt

Contingency: Suspend inappropriate role; review affected orders

Status: Open; illustrative assessment

5. RISK-005

ID: RISK-005

Risk: Payment handling clears wrong balances

Owner: Finance

Likelihood: 3

Impact: 4

Score: 12

Mitigation: Reference checking; no automatic write-off; reconcile allocation totals

Trigger: Unmatched/overallocated receipts

Contingency: Hold for manual investigation

Status: Open; illustrative assessment

6. RISK-006

ID: RISK-006

Risk: Reporting duplicates values through child joins

Owner: IT

Likelihood: 3

Impact: 4

Score: 12

Mitigation: Grain contract, fixture oracle and independent SQL checks

Trigger: Totals change when payment rows added

Contingency: Withdraw report and reconcile aggregates

Status: Open; illustrative assessment

7. RISK-007

ID: RISK-007

Risk: Desk-based SAP fit differs from tenant behaviour

Owner: IT

Likelihood: 4

Impact: 4

Score: 16

Mitigation: Validate release, scope, entitlement and configuration in future sandbox

Trigger: Unverified control or unsupported field

Contingency: Retain manual control until demonstrated

Status: Open; illustrative assessment

8. RISK-008

ID: RISK-008

Risk: Users continue parallel spreadsheets

Owner: Customer Service

Likelihood: 3

Impact: 3

Score: 9

Mitigation: Role training; one exception owner; feedback and retirement plan

Trigger: Conflicting status reports

Contingency: Reconcile and identify authoritative record

Status: Open; illustrative assessment

9. RISK-009

ID: RISK-009

Risk: Benefit case mistakes capacity for cash

Owner: Finance

Likelihood: 3

Impact: 4

Score: 12

Mitigation: Separate capacity, cash and working-capital assumptions

Trigger: Unsupported ROI claim

Contingency: Use sensitivity analysis; defer investment decision

Status: Open; illustrative assessment

10. RISK-010

ID: RISK-010

Risk: Portfolio overstates learner experience or readiness

Owner: Learner

Likelihood: 3

Impact: 5

Score: 15

Mitigation: Label simulation/AI assistance; oral review before resume use

Trigger: Cannot explain decisions independently

Contingency: Revise claims; repeat learning tasks

Status: Open; illustrative assessment

Appendix — Change and training

Editable source: phase-10/change-impact.csv

1. Sales

Role: Sales

Change: Promise only checked quantity/date; retain original promise; correction queue

Training: 45-minute NS-ORD-1001 and missing-data exercise

Communication: Before trial: explain why checks protect promise quality

Support: Sales key user → process owner → IT if technical

2. Warehouse

Role: Warehouse

Change: Separate physical dispatch from posting; resolve quantity mismatch

Training: 45-minute partial pick and delayed-posting exercise

Communication: Daily handoff checklist and unresolved stock review

Support: Warehouse key user → Warehouse owner → IT

3. Finance

Role: Finance

Change: Own billable population; controlled correction and payment handling

Training: 60-minute billing retry, credit release and partial-payment cases

Communication: Daily unbilled/credit/payment review; weekly control issues

Support: Finance key user → Finance owner → IT

4. Customer Service

Role: Customer Service

Change: Use authoritative status; capture partial agreement and next action

Training: 30-minute status enquiry and customer agreement role-play

Communication: Short quick guide; weekly feedback session in pilot

Support: Customer Service key user → relevant process owner

5. IT

Role: IT

Change: Support role matrix, data contracts, refresh and evidence retention

Training: 60-minute access, failed-refresh and reconciliation exercise

Communication: Release notes and change-impact review before changes

Support: IT support → authorized platform support where applicable

Appendix — Business-case options

Editable source: phase-10/business-case-options.csv

1. A: Improve manual controls

Option: A: Improve manual controls

One off GBP: 2000

Annual running GBP: 500

Minutes saved per order: 2

Annual orders assumed: 6000

Loaded hourly rate GBP: 25

Annual capacity hours: 200.0

Capacity equivalent GBP: 5000.0

Cash savings GBP: Unproven

Cost basis: Illustrative planning placeholder, not supplier quote or SAP price

2. B: Standardise process and shared reporting

Option: B: Standardise process and shared reporting

One off GBP: 8000

Annual running GBP: 1500

Minutes saved per order: 5

Annual orders assumed: 6000

Loaded hourly rate GBP: 25

Annual capacity hours: 500.0

Capacity equivalent GBP: 12500.0

Cash savings GBP: Unproven

Cost basis: Illustrative planning placeholder, not supplier quote or SAP price

3. C: SAP-aligned future state

Option: C: SAP-aligned future state

One off GBP: 60000

Annual running GBP: 18000

Minutes saved per order: 8

Annual orders assumed: 6000

Loaded hourly rate GBP: 25

Annual capacity hours: 800.0

Capacity equivalent GBP: 20000.0

Cash savings GBP: Unproven

Cost basis: Illustrative planning placeholder, not supplier quote or SAP price

Appendix — Sensitivity

Editable source: phase-10/sensitivity.csv

1. B sensitivity

Option: B sensitivity

Annual orders: 6000

Minutes saved: 2

Hourly rate GBP: 20

Capacity hours: 200.0

Capacity equivalent GBP: 4000.0

Annual running GBP: 1500

Interpretation: Capacity valuation only; not cash ROI

2. B sensitivity

Option: B sensitivity

Annual orders: 6000

Minutes saved: 2

Hourly rate GBP: 25

Capacity hours: 200.0

Capacity equivalent GBP: 5000.0

Annual running GBP: 1500

Interpretation: Capacity valuation only; not cash ROI

3. B sensitivity

Option: B sensitivity

Annual orders: 6000

Minutes saved: 2

Hourly rate GBP: 30

Capacity hours: 200.0

Capacity equivalent GBP: 6000.0

Annual running GBP: 1500

Interpretation: Capacity valuation only; not cash ROI

4. B sensitivity

Option: B sensitivity

Annual orders: 6000

Minutes saved: 5

Hourly rate GBP: 20

Capacity hours: 500.0

Capacity equivalent GBP: 10000.0

Annual running GBP: 1500

Interpretation: Capacity valuation only; not cash ROI

5. B sensitivity

Option: B sensitivity

Annual orders: 6000

Minutes saved: 5

Hourly rate GBP: 25

Capacity hours: 500.0

Capacity equivalent GBP: 12500.0

Annual running GBP: 1500

Interpretation: Capacity valuation only; not cash ROI

6. B sensitivity

Option: B sensitivity

Annual orders: 6000

Minutes saved: 5

Hourly rate GBP: 30

Capacity hours: 500.0

Capacity equivalent GBP: 15000.0

Annual running GBP: 1500

Interpretation: Capacity valuation only; not cash ROI

7. B sensitivity

Option: B sensitivity

Annual orders: 6000

Minutes saved: 8

Hourly rate GBP: 20

Capacity hours: 800.0

Capacity equivalent GBP: 16000.0

Annual running GBP: 1500

Interpretation: Capacity valuation only; not cash ROI

8. B sensitivity

Option: B sensitivity

Annual orders: 6000

Minutes saved: 8

Hourly rate GBP: 25

Capacity hours: 800.0

Capacity equivalent GBP: 20000.0

Annual running GBP: 1500

Interpretation: Capacity valuation only; not cash ROI

9. B sensitivity

Option: B sensitivity

Annual orders: 6000

Minutes saved: 8

Hourly rate GBP: 30

Capacity hours: 800.0

Capacity equivalent GBP: 24000.0

Annual running GBP: 1500

Interpretation: Capacity valuation only; not cash ROI

Appendix — Executed data checks

Editable source: phase-08/data/data_checks.csv

1. DQ-001

check id: DQ-001

name: customers primary key

status: Passed

detail: 20 unique non-null keys

execution type: Executed Python/SQLite data check

date: 2026-10-11

2. DQ-002

check id: DQ-002

name: products primary key

status: Passed

detail: 5 unique non-null keys

execution type: Executed Python/SQLite data check

date: 2026-10-11

3. DQ-003

check id: DQ-003

name: orders primary key

status: Passed

detail: 300 unique non-null keys

execution type: Executed Python/SQLite data check

date: 2026-10-11

4. DQ-004

check id: DQ-004

name: deliveries primary key

status: Passed

detail: 285 unique non-null keys

execution type: Executed Python/SQLite data check

date: 2026-10-11

5. DQ-005

check id: DQ-005

name: invoices primary key

status: Passed

detail: 262 unique non-null keys

execution type: Executed Python/SQLite data check

date: 2026-10-11

6. DQ-006

check id: DQ-006

name: receipts primary key

status: Passed

detail: 209 unique non-null keys

execution type: Executed Python/SQLite data check

date: 2026-10-11

7. DQ-007

check id: DQ-007

name: allocations primary key

status: Passed

detail: 197 unique non-null keys

execution type: Executed Python/SQLite data check

date: 2026-10-11

8. DQ-008

check id: DQ-008

name: events primary key

status: Passed

detail: 688 unique non-null keys

execution type: Executed Python/SQLite data check

date: 2026-10-11

9. DQ-009

check id: DQ-009

name: Order master references

status: Passed

detail: Customer and product parents exist

execution type: Executed Python/SQLite data check

date: 2026-10-11

10. DQ-010

check id: DQ-010

name: Delivery/invoice parent references

status: Passed

detail: No orphan fulfilment records

execution type: Executed Python/SQLite data check

date: 2026-10-11

11. DQ-011

check id: DQ-011

name: Receipt/allocation references

status: Passed

detail: No orphan payment links

execution type: Executed Python/SQLite data check

date: 2026-10-11

12. DQ-012

check id: DQ-012

name: Receipt customer references

status: Passed

detail: Every cash receipt has a known synthetic customer

execution type: Executed Python/SQLite data check

date: 2026-10-11

13. DQ-013

check id: DQ-013

name: Order value arithmetic

status: Passed

detail: Integer pence, tax excluded

execution type: Executed Python/SQLite data check

date: 2026-10-11

14. DQ-014

check id: DQ-014

name: Quantity bounds and cancelled fulfilment

status: Passed

detail: No overdelivery or cancelled order shipment

execution type: Executed Python/SQLite data check

date: 2026-10-11

15. DQ-015

check id: DQ-015

name: Partial permission

status: Passed

detail: Split/short shipments require permission in synthetic baseline

execution type: Executed Python/SQLite data check

date: 2026-10-11

16. DQ-016

check id: DQ-016

name: Delivery chronology

status: Passed

detail: Physical and posting events separated; future actuals excluded

execution type: Executed Python/SQLite data check

date: 2026-10-11

17. DQ-017

check id: DQ-017

name: Invoice chronology and amount

status: Passed

detail: Only observed goods-issued deliveries billed; one active invoice per delivery

execution type: Executed Python/SQLite data check

date: 2026-10-11

18. DQ-018

check id: DQ-018

name: No duplicate active invoice

status: Passed

detail: Simplified dataset excludes correction documents

execution type: Executed Python/SQLite data check

date: 2026-10-11

19. DQ-019

check id: DQ-019

name: Allocation upper bounds

status: Passed

detail: No cash or invoice over-allocation

execution type: Executed Python/SQLite data check

date: 2026-10-11

20. DQ-020

check id: DQ-020

name: Allocation chronology/customer

status: Passed

detail: Matching customers and valid event order

execution type: Executed Python/SQLite data check

date: 2026-10-11

21. DQ-021

check id: DQ-021

name: Python versus independent SQL KPIs

status: Passed

detail: All six calculations agree before rounding

execution type: Executed Python/SQLite data check

date: 2026-10-11

22. DQ-022

check id: DQ-022

name: Example retained

status: Passed

detail: Representative case retains unresolved availability; no fabricated fulfilment

execution type: Executed Python/SQLite data check

date: 2026-10-11

23. DQ-023

check id: DQ-023

name: Split and partial payment coverage

status: Passed

detail: Dataset includes split deliveries and unpaid residual analytical amounts

execution type: Executed Python/SQLite data check

date: 2026-10-11

24. DQ-024

check id: DQ-024

name: Four-order hand oracle

status: Passed

detail: Cycle 4d; on-time 50%; fulfilment 46.6667%; error 25%; manual 50%; billing 2d; value £10,000

execution type: Executed Python/SQLite data check

date: 2026-10-11

Appendix — Executed catalogue checks

Editable source: quality/consistency-checks.csv

1. QC-001

id: QC-001

check: Catalogue unique IDs

status: Passed

detail: 3 BR + 15 FR + 5 NFR; 15 stories; 22 test designs

type: Executed artifact consistency check

2. QC-002

id: QC-002

check: All three business problems traced

status: Passed

detail: Every supplied problem has a trace row

type: Executed artifact consistency check

3. QC-003

id: QC-003

check: Every functional requirement traced

status: Passed

detail: No functional requirement missing from trace

type: Executed artifact consistency check

4. QC-004

id: QC-004

check: Story/test references valid

status: Passed

detail: All story requirement and test references resolve

type: Executed artifact consistency check

5. QC-005

id: QC-005

check: Trace and test bidirectional

status: Passed

detail: Trace test links include the same requirement

type: Executed artifact consistency check

6. QC-006

id: QC-006

check: All tests link valid requirements

status: Passed

detail: No dangling test requirements

type: Executed artifact consistency check

7. QC-007

id: QC-007

check: All NFRs covered

status: Passed

detail: All five NFRs linked to planned cases

type: Executed artifact consistency check

8. QC-008

id: QC-008

check: SAP execution labels honest

status: Passed

detail: No invented test result or SAP defect

type: Executed artifact consistency check

9. QC-009

id: QC-009

check: Seven exceptions covered

status: Passed

detail: Includes positive and negative partial-delivery permission; pre/post dispatch cancellation

type: Executed artifact consistency check

10. QC-010

id: QC-010

check: Rules and process change IDs

status: Passed

detail: All 12 rules referenced; 15 unique changes

type: Executed artifact consistency check

11. QC-011

id: QC-011

check: Fit assessment covers FRs

status: Passed

detail: 15 desk-based fit assessments; no claimed tenant demo

type: Executed artifact consistency check

12. QC-012

id: QC-012

check: Source evidence completeness

status: Passed

detail: Official URLs/access date and demonstration boundary recorded

type: Executed artifact consistency check

13. QC-013

id: QC-013

check: Unverified sources explicit

status: Passed

detail: Two unreadable Help pages not treated as verified capability

type: Executed artifact consistency check

14. QC-014

id: QC-014

check: Synthetic data checks passed

status: Passed

detail: 24 checks incl. independent Python/SQL and hand-calculated fixture

type: Executed artifact consistency check

15. QC-015

id: QC-015

check: Required files present

status: Passed

detail: Core narratives/maps exist

type: Executed artifact consistency check

16. QC-016

id: QC-016

check: Learning questions retained

status: Passed

detail: All twelve learning reviews present; none marked passed

type: Executed artifact consistency check